AI in Medicine I

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1 Introduction to AI for Health

1.1 From expert systems to foundation models

Expert systems (1970s). An **expert system** encodes medical knowledge by hand: a **knowledge base** of facts and rules, an **inference engine** that applies them to a new case, and a user interface through which the physician describes the case and receives advice with an explanation. **MYCIN** identified the bacteria causing severe infections and recommended antibiotics with a dosage adjusted to body weight, performing better than infectious-disease experts.

Probabilistic models (1980s). **INTERNIST-1/QMR** supports diagnosis in internal medicine with a probabilistic model: a bipartite directed graph of binary disease and symptom variables, with an edge from each disease to the symptoms it causes (fig. 1); diagnosis means inferring the diseases from the observed symptoms. The graph was elicited from doctors with years of manual work, and it drove advances in Bayesian networks and approximate inference. Its limits are those of every hand-built system: symptoms must be entered manually, and the model is hard to maintain and to generalise.

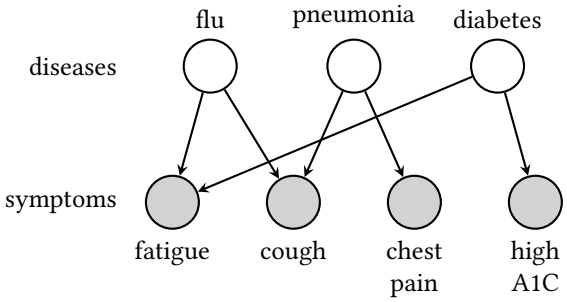


Figure 1: A small piece of a disease-symptom network in the style of INTERNIST-1/QMR.

Foundation models (2020s). A **foundation model** for **generalist medical AI** (GMAI) is trained by self-supervision on many modalities at once (images, electronic health records (EHRs), signals, omics, graphs, text, audio), uses medical knowledge (literature, clinical notes, knowledge graphs), and supports flexible interactions: the task is specified dynamically, e.g. by a question, with multimodal inputs and outputs. Examples are bedside decision support (an alert that the model explains on request), radiology reports whose findings are grounded in image regions, and assistance during procedures such as endoscopy. Deployment requires approval, validation, audits and analyses of bias and fairness.

Large language models. An LLM such as ChatGPT is a GPT model pretrained on large text corpora, refined by **reinforcement learning from human feedback** (RLHF) and wrapped in **guardrails** for safety. In medicine it can serve patient care (documentation, summarisation, medical knowledge), education and research.

1.2 Medical imaging

The goal is a model that integrates all data about a patient (images, vital signs, lab tests, genomics and other omics, wearables, clinical records and history, lifestyle) and outputs a diagnosis or prognosis. Imaging visualises organs and tissue inside the body: an **imaging modality** produces a map of some physical property of the tissue.

Definition 1.1 (Image)**.** An **image** is a function of the position: $f(x, y)$ in 2D, $f(x, y, z)$ in 3D, or $f(x, y, z, t)$ in 4D (3D over time). Its values are scalars, displayed as intensity, or vectors, which are harder to visualise.

Modalities differ in the physical mechanism that generates contrast (radiation, magnetisation, sound, light) and in their risks: some use harmful **ionising radiation**, some are **invasive** (a contrast agent is injected or a device inserted into the body), many are harmless and non-invasive.

- **X-ray radiography**: ionising radiation; a 2D projection of the 3D body.
- **Computed tomography** (CT): X-rays restricted by a collimator to one or a few slices; source and detector rotate around the patient, and the slice image $I(x, y) = \mu(x, y, z_0)$ (μ the X-ray attenuation) is computed from the projections taken at many angles. Stacked slices give a 3D volume.
- **Magnetic resonance imaging** (MRI): by varying the acquisition it maps different tissue properties of the same anatomy (**T1-weighted**, **T2-weighted**, **proton density** images).
- **Ultrasound** (US): images flow via the Doppler effect; its main noise is **speckle**, which depends on the signal.
- **SPECT** and **PET**: nuclear imaging of an injected radioactive tracer; functional, low resolution.

Anatomical imaging shows structure, **functional imaging** shows physiological activity; hybrid modalities such as **PET/CT** overlay function (PET) on anatomy (CT), as in fig. 2. Table 1 summarises the properties.

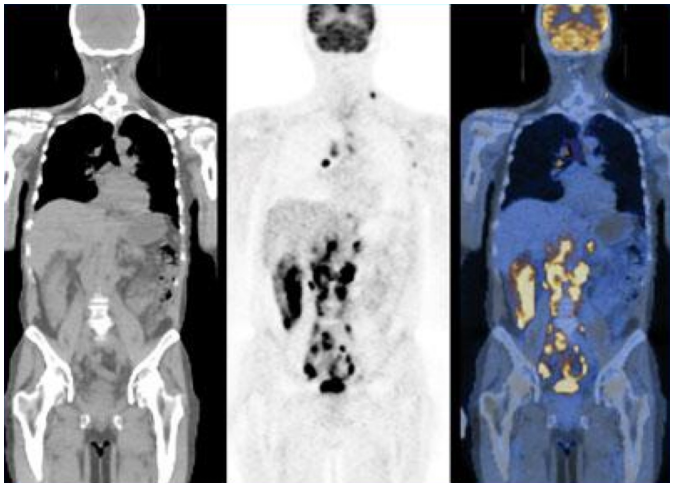


Figure 2: Whole-body PET/CT: CT (anatomy), PET (function, dark spots of high tracer uptake), and their overlay.

	2D	3D	4D	Ionising	Invasive	Anatomical	Functional	High res.
X-ray	✓	×	×	✓	×	✓	×	✓
Ultrasound	✓	✓	✓	×	×	✓	✓	✓
CT	n/a	✓	✓	✓	×	✓	×	✓
MRI	✓	✓	✓	×	×	✓	✓	✓
SPECT	n/a	✓	×	✓	✓	×	✓	×
PET	n/a	✓	✓	✓	✓	×	✓	×

Table 1: Comparison of imaging modalities.

1.3 Signals and clinical records

Signals. Wearables, **electroencephalography** (EEG, electrical activity of the brain) and **electrocardiography** (ECG, of the heart) produce multichannel time series, represented in the time domain, the frequency domain (spectral representation) or a graph domain (state representation). For example, an **inertial measurement unit** (IMU) worn on the body records velocity, acceleration and displacement, and a classifier maps them to an activity (walk, fall, stand, sit); for EEG and ECG a typical task is to

detect abnormal recordings.

Records and standards. Large public datasets combine these sources: **MIMIC** holds de-identified intensive-care records (vital-sign time series, lab values, medication, images, free-text reports), and **UK Biobank** is a population cohort with imaging, genetics, lifestyle and clinical data. Standard codes make records comparable and machine-readable:

- **ICD-10** (International Classification of Diseases) codes diseases hierarchically: a letter and two digits give the category (the letter range gives the chapter, e.g. neoplasms or injuries), and further characters encode etiology, location, laterality and extension;
- **LOINC** identifies laboratory observations, **NDC** (National Drug Codes) identifies drugs;
- **UMLS** (Unified Medical Language System) maps between medical vocabularies, a thesaurus and ontology of biomedical concepts that enables natural language processing.

2 Basic Deep Learning Architectures

Typical imaging tasks are **image classification** (a categorical label: modality such as MR, CT or angiography, MR sequence such as T1, T2 or PD, image quality pass/fail, disease state such as Alzheimer's disease versus normal), **image regression** (a quantity: the patient's age, a quality score in $[0, 1]$) and **region classification** (classifying sub-regions of an image: anomaly detection, anatomy or organ detection). The classical pipeline first computes hand-crafted features $\boldsymbol{x} = f(\mathbf{I})$ of the image $\mathbf{I}$ (raw pixel values, SIFT, HoG, textons) and then trains a model $h_\Theta(\boldsymbol{x})$ on them; a deep network learns the whole map $h_\Theta(\mathbf{I})$ from the image to the prediction, features included.

2.1 Convolutional neural networks

Neurons. A neuron computes a weighted sum of its inputs and applies a non-linear **activation function** f_a (fig. 3):

$$\hat{y}(\boldsymbol{x}) = f_a\Big(\sum_{i=1}^{N} x_i\theta_i + \theta_0\Big),$$

where the bias θ_0 is the weight of a constant input $x_0 = 1$. Common activations are the sigmoid $S(x) = 1/(1 + e^{-x})$ with values in $(0, 1)$, $\tanh(x) = (e^x - e^{-x})/(e^x + e^{-x})$ with values in $(-1, 1)$, and the rectifier (**ReLU**) $f(x) = \max(0, x)$. Training minimises a loss over the data $(\mathbf{X}, \mathbf{Y})$, e.g. the mean squared error, by gradient descent with step size $\eta > 0$:

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2, \qquad \theta^* = \arg\min_\theta \mathcal{L}(\theta, \mathbf{X}, \mathbf{Y}), \qquad \theta_i^{t+1} = \theta_i^t - \eta\,\frac{\partial \mathcal{L}}{\partial \theta_i^t}.$$

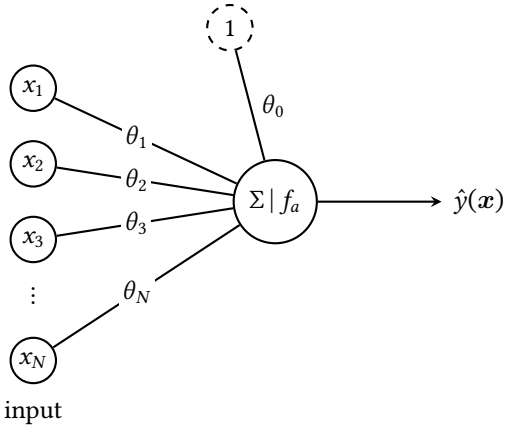


Figure 3: A neuron: weighted sum of the inputs plus bias, followed by the activation f_a.

From fully connected to convolutional layers. In a fully connected network each neuron of the first layer sees every pixel and detects one pattern, so it needs one weight per pixel: $512^2 = 262{,}144$ weights for a 512×512 image. Two properties of images reduce this (fig. 4).

- **Locality**: features are local, only neighbouring pixels are correlated. In a **locally connected network** each neuron sees only a small window, its **kernel** (a 10×10 kernel has 100 weights), but it detects its pattern only at its own location.
- **Stationarity**: the same feature may appear anywhere in the image. So all neurons share the same kernel weights (**weight sharing**), and sliding one kernel over the image is a convolution.

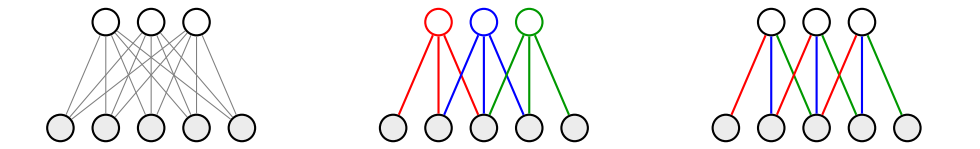


Figure 4: From fully connected to convolution (1D, kernel of size 3); colours mark distinct weights.

The single neuron $a(\boldsymbol{x}) = f_a(\sum_i \theta_i x_i + b)$ thus becomes a **feature map** $\mathbf{a}(\boldsymbol{x}) = f_a(\boldsymbol{\theta} * \boldsymbol{x} + b)$, which is itself an image.

Definition 2.1 (2D convolution)**.** For an image f and a kernel g of size $n \times n$ (indices $0, \dots, n-1$),

$$(f * g)[x, y] = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} f\left(x - i + \left\lfloor \frac{n}{2} \right\rfloor,\ y - j + \left\lfloor \frac{n}{2} \right\rfloor\right) g(i, j);$$

the shift by $\lfloor n/2 \rfloor$ centres the kernel on the pixel (x, y).

Fixed kernels extract simple features: horizontal edges, vertical edges (the transpose) and, with the Laplacian, edges in every direction,

$$\begin{pmatrix} -1 & -1 & -1 \\ 1 & 1 & 1 \\ -1 & -1 & -1 \end{pmatrix}, \qquad \begin{pmatrix} -1 & 1 & -1 \\ -1 & 1 & -1 \\ -1 & 1 & -1 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

A CNN learns its kernels instead.

Convolutional layers. A **convolutional layer** learns D kernels at once and stacks the D resulting feature maps into a D-channel image (like the RGB channels of a colour image). If its input has C channels, each kernel extends over the full depth: it has size $k_1 \times k_2 \times C$ and sums over the channels (fig. 5). Deeper layers process the feature maps of the previous layer and combine them into more complex representations. A layer has

$$k_1 k_2 C D \text{ weights (plus } D \text{ biases)},$$

e.g. $5 \cdot 5 \cdot 3 \cdot 4 = 300$ weights for $D = 4$ filters of size 5×5 on an RGB input ($C = 3$). The count does not depend on the image size, and each weight is used, hence trained, at every position, so it sees much more training data.

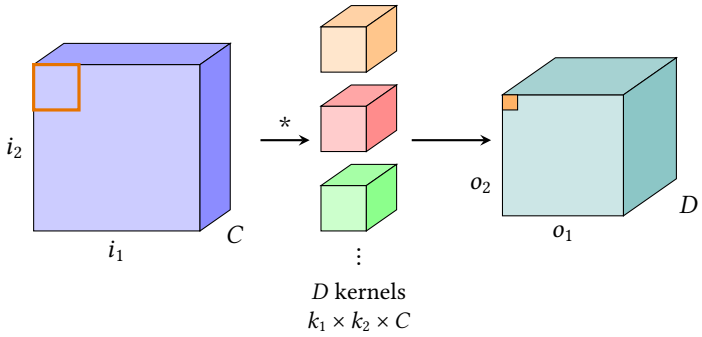


Figure 5: A convolutional layer: each of the D kernels spans all C input channels and produces one output feature map.

Definition 2.2 (Output size of a convolution)**.** Along axis j let i_j be the input size, k_j the kernel size, p_j the **zero padding** (zeros added at both the beginning and the end of the axis) and s_j the **stride** (distance between consecutive kernel positions). The feature map has size

$$o_j = \left\lfloor \frac{i_j - k_j + 2p_j}{s_j} \right\rfloor + 1.$$

Indeed the padded axis has length $i_j + 2p_j$, and the kernel fits at the positions $0, s_j, 2s_j, \dots, ms_j$ with $ms_j + k_j \le i_j + 2p_j$, so $m = \lfloor (i_j + 2p_j - k_j)/s_j \rfloor$ and $o_j = m + 1$. Without padding and stride ($p_j = 0$, $s_j = 1$) this is $o_j = i_j - k_j + 1$ (e.g. $5 - 3 + 1 = 3$); with padding only, $o_j = i_j - k_j + 2p_j + 1$ (e.g. $i_j = 5$, $k_j = 4$, $p_j = 2$ give 6); fig. 6 shows a case with stride.

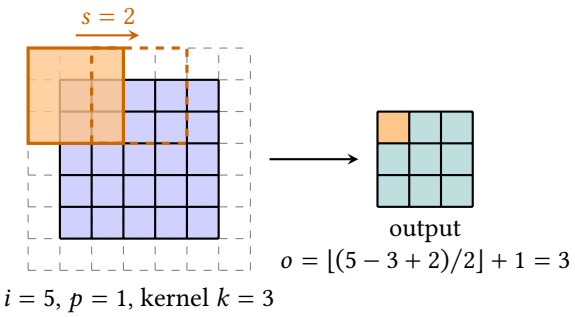


Figure 6: Convolution with padding and stride: the kernel jumps by $s = 2$ over the zero-padded input.

Receptive field. The **receptive field** of an output pixel is the input region it depends on; it grows with depth. A $k \times k$ convolution without padding and with stride 1 shrinks the map by $k - 1$ and enlarges the receptive field by $k - 1$, so after L such layers the receptive field is $L(k - 1) + 1$ (fig. 7). For instance four 5×5 layers map a 17×17 input to 13, 9, 5 and finally 1 pixel, whose receptive field is the whole 17×17 input. Small filters in many layers use fewer parameters than one large filter: $4 \cdot 25 = 100$ against $17^2 = 289$ weights (per pair of input and output channels). Without padding the feature maps shrink with depth.

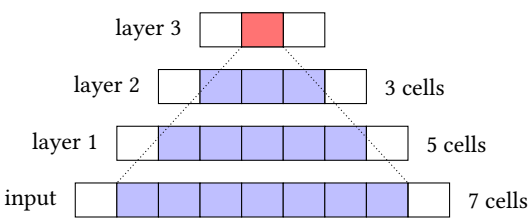


Figure 7: Receptive field in 1D: three convolutions with $k = 3$; the red pixel sees 3, 5, 7 cells below it.

Pooling. Downsampling reduces the size (memory) of deeper layers, gives invariance to small translations, and the contraction forces the network to learn high-level features. A **pooling** operator has a kernel size k and a stride s (the output size follows the same formula as a convolution) and is applied to each feature map separately, so the number of channels is unchanged: e.g. $224 \times 224 \times 10$ becomes $112 \times 112 \times 10$ with $k = s = 2$. Its kernel function, e.g. **max pooling** or **mean pooling**, has no parameters (fig. 8). Replacing the fixed function by learnable weights $w_1, \dots, w_{k^2}$ turns pooling into a strided convolution, i.e. a learned downsampling.

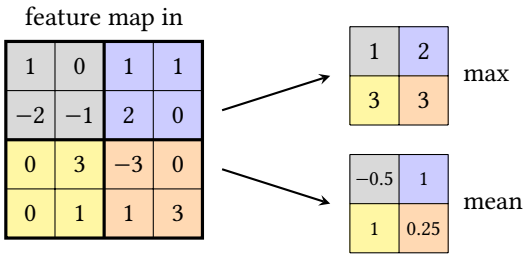


Figure 8: Max and mean pooling with $k = s = 2$: each coloured 2×2 block gives one output value.

Upsampling. To enlarge a feature map by a factor s, insert $s - 1$ zeros between neighbouring entries, pad, and convolve. With a fixed kernel this is interpolation: the kernel with rows $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$, $(\frac{1}{2}, 1, \frac{1}{2})$, $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ copies the original values and fills each new entry with the mean of its two or four original neighbours, i.e. bilinear interpolation (fig. 9). A **transposed convolution** (or **deconvolution**) learns this kernel, a data-driven upsampling. Its kernel size k sets how much context influences each output value, its stride s is the upsampling factor, and its padding p must be chosen carefully to obtain exactly the wanted output shape. Equivalently, a deconvolution with (k, s, p) goes back from the output shape of the strided convolution with the same (k, s, p) to that convolution's input shape.

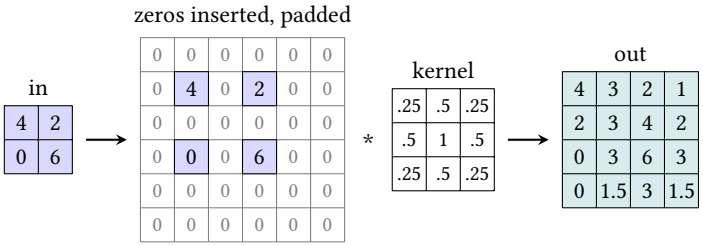


Figure 9: Upsampling a 2×2 map with $s = 2$: zero insertion, then a 3×3 convolution (here the fixed bilinear kernel).

2.2 Classic CNN architectures

LeNet. The first successful CNN on a real-world problem (handwritten digit recognition) set the pattern later architectures follow: a **feature-learning** part of alternating convolution and subsampling layers, then a **classification** part of fully connected layers (an MLP), as in fig. 10. It subsamples by average pooling, uses tanh activations, and connects two of its convolutional stages sparsely (each feature map sees only a subset of the previous maps).

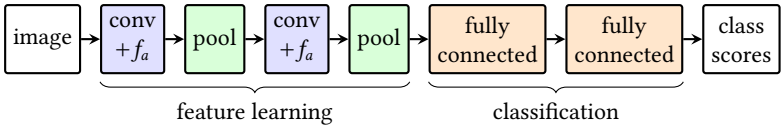


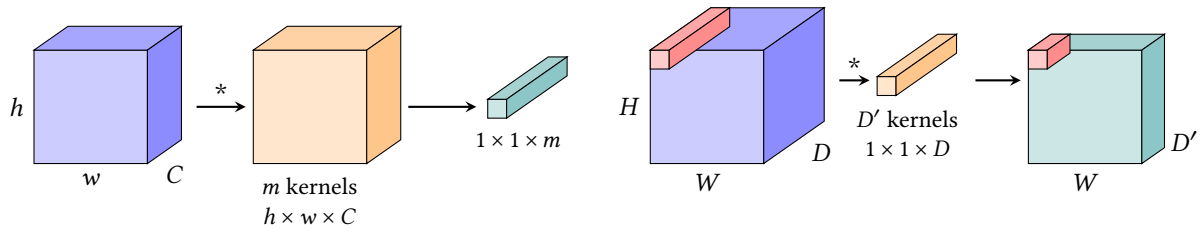
Figure 10: Basic CNN pattern: convolution and pooling learn features, fully connected layers classify.

AlexNet. A much larger LeNet whose novelties are ReLU activations and max pooling, convolutional layers stacked directly on top of each other (instead of a pooling layer after every convolution), **dropout** regularisation, **data augmentation** (random transformations and intensity variations) and training on GPUs. Its first-layer kernels turn out to be oriented edge detectors and colour blobs. More generally, features get more abstract with depth: edges and blobs in the first layer, textures in the middle, object parts in deeper layers, object classes at the output.

VGGNet. Studies the effect of depth with a uniform design: only 3×3 convolutions with stride 1 and 2×2 max pooling, followed by three fully connected layers. It is very large, about 138 million parameters, and roughly 103 million of them sit in the first fully connected layer alone ($7 \cdot 7 \cdot 512$ inputs times 4096 units): the size of CNNs comes mostly from their fully connected layers.

Fully convolutional networks. A fully connected neuron on C feature maps of size $h \times w$ computes a weighted sum of all hwC activations plus a bias, which is exactly a convolution with one $h \times w \times C$ kernel without padding, producing a 1×1 output (fig. 11a); m such neurons are m kernels, i.e. m feature maps of size 1×1. Replacing every fully connected layer in this way gives a **fully convolutional network**, which accepts inputs of any size: a larger input yields larger feature maps and a map of predictions instead of a single one. The receptive field of each prediction stays the same, and each prediction depends only on its own receptive field, not on the whole input. With the four 5×5 layers above, a 25×25 input gives a 9×9 output ($25 - 4 \cdot 4 = 9$).

1×1 **convolutions.** **Network in network** inserts small MLPs (fully connected layers) between convolutional layers, applied at every position; this is the same as a 1×1 convolution. A 1×1 convolution with D' filters maps an $H \times W \times D$ volume to $H \times W \times D'$: each output value $f_a\left(\sum_{c=1}^{D} w_c \, a_c(x, y) + b\right)$ combines the D channels at the same pixel (fig. 11b). With $D' < D$ it reduces the number of channels cheaply.



(a) fully connected layer = convolution

(b) 1×1 convolution, $D' < D$

Figure 11: (a) A fully connected layer is a convolution with kernels as large as the input. (b) A 1×1 convolution maps the D values of a pixel to D' values.

GoogLeNet (Inception v1). A deeper network with **global average pooling** (each final feature map averaged to one value) as its final layer, and **auxiliary classifiers** attached to intermediate layers, used only during training. It is built from **Inception modules** (fig. 12), derived from the network-in-network idea: 1×1, 3×3 and 5×5 convolutions and 3×3 max pooling run in parallel on the same input and their outputs are concatenated along the channels (split-transform-merge), so the network *learns* which filter size to use. The 1×1 convolutions before the large kernels and after the pooling are **bottleneck** layers that reduce the number of channels; the module thus has the representational power of large dense layers with far fewer parameters and operations.

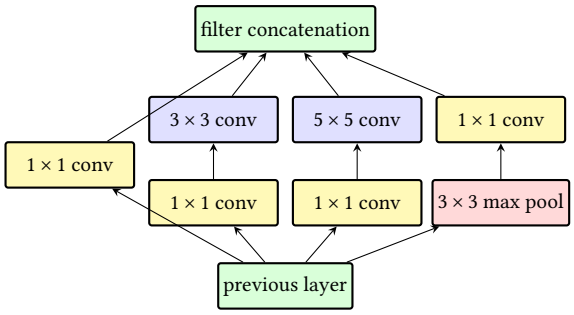


Figure 12: Inception module: parallel branches with 1×1 bottlenecks, concatenated along the channels.

Example 2.3 (Cost of a bottleneck)**.** Map a $14 \times 14 \times 480$ volume to $14 \times 14 \times 48$ with 5×5 convolutions (size-preserving padding). Each output value needs $5 \cdot 5 \cdot 480$ multiplications, so directly

$$(14 \cdot 14 \cdot 48)\,(5 \cdot 5 \cdot 480) \approx 112.9\text{M operations.}$$

With a 1×1 convolution to 16 channels first, the cost is $(14 \cdot 14 \cdot 16)(1 \cdot 1 \cdot 480) \approx 1.5$M for the 1×1 layer plus $(14 \cdot 14 \cdot 48)(5 \cdot 5 \cdot 16) \approx 3.8$M for the 5×5 layer, 5.3M in total: about 21 times fewer.

Inception v2 and v3. Inception v2 factorises large convolutions. A 5×5 (or 7×7) convolution is replaced by a stack of 3×3 ones: two 3×3 layers have a 5×5 receptive field with $2 \cdot 9 = 18$ instead of 25 weights. An $n \times n$ convolution is replaced by a $1 \times n$ followed by an $n \times 1$ one ($2n$ instead of n^2 weights). Grid-size reduction is done efficiently by parallel stride-2 convolution and stride-2 pooling branches whose outputs are concatenated.

Inception v3 adds the RMSProp optimiser, factorised 7×7 convolutions, batch normalisation in the auxiliary classifiers and **label smoothing**, a regularisation added to the loss that copes with label noise and prevents the network from becoming too confident about one class, which reduces overfitting. Even deeper networks (ResNet) belong to the advanced architectures.

2.3 Evaluating classifiers and case studies

Precision, accuracy, robustness. For a measurement, **precision** describes the random errors (statistical variability): the repeatability or reproducibility of the measurement. **Accuracy** describes the systematic errors (statistical bias), the difference between the result and the true value, which ISO calls *trueness*; ISO alternatively defines accuracy as the combination of both random and systematic errors, so that high accuracy requires both high precision and high trueness. **Robustness** is how much performance degrades as noise levels or imaging artefacts vary. (Precision as a classification metric, below, is a different quantity.)

Confusion matrix. For a binary classifier, compare the actual condition with the predicted one:

	predicted positive (PP)	predicted negative (PN)
actual positive (P)	true positive (TP), hit	false negative (FN), miss, type II error
actual negative (N)	false positive (FP), false alarm, type I error	true negative (TN), correct rejection

so $\mathrm{P} = \mathrm{TP} + \mathrm{FN}$, $\mathrm{N} = \mathrm{TN} + \mathrm{FP}$, $\mathrm{PP} = \mathrm{TP} + \mathrm{FP}$, $\mathrm{PN} = \mathrm{FN} + \mathrm{TN}$.

Definition 2.4 (Basic metrics)**.**

$$\mathrm{ACC} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{P}+\mathrm{N}}, \qquad \mathrm{PPV} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \qquad \mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{P}} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \qquad \mathrm{TNR} = \frac{\mathrm{TN}}{\mathrm{N}} = \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}}.$$

Accuracy (ACC) is the fraction of correct decisions; **precision** or **positive predictive value** (PPV) answers "how many selected items are relevant?"; **recall**, **sensitivity**, **hit rate** or **true positive rate**

(TPR) answers "how many relevant items are selected?"; **specificity** or **true negative rate** (TNR) is the analogue for the negatives.

Other common metrics are the **false positive rate** $\mathrm{FPR} = \mathrm{FP}/\mathrm{N} = 1 - \mathrm{TNR}$, the **false negative rate** $\mathrm{FNR} = \mathrm{FN}/\mathrm{P} = 1 - \mathrm{TPR}$, the **negative predictive value** $\mathrm{NPV} = \mathrm{TN}/\mathrm{PN}$, the **balanced accuracy** $(\mathrm{TPR} + \mathrm{TNR})/2$, and the F_1 **score**, the harmonic mean of precision and recall; with $\mathrm{PPV} = \mathrm{TP}/\mathrm{PP}$ and $\mathrm{TPR} = \mathrm{TP}/\mathrm{P}$,

$$F_1 = \frac{2\,\mathrm{PPV}\cdot\mathrm{TPR}}{\mathrm{PPV}+\mathrm{TPR}} = \frac{2\,\mathrm{TP}^2/(\mathrm{PP}\cdot\mathrm{P})}{\mathrm{TP}\,(\mathrm{P}+\mathrm{PP})/(\mathrm{PP}\cdot\mathrm{P})} = \frac{2\,\mathrm{TP}}{\mathrm{P}+\mathrm{PP}} = \frac{2\,\mathrm{TP}}{2\,\mathrm{TP}+\mathrm{FP}+\mathrm{FN}}.$$

ROC curve and AUC. A classifier that outputs a score decides by comparing it with a threshold, and every threshold gives one point $(\mathrm{FPR}, \mathrm{TPR})$. The **receiver operating characteristic** (ROC) curve traces these points as the threshold varies (fig. 13). The perfect classifier sits at $(0, 1)$, random guessing gives points on the diagonal $\mathrm{TPR} = \mathrm{FPR}$, and curves closer to the top-left corner are better. The **area under the ROC curve** (AUC) lies in $[0, 1]$: it is 0 for a model whose predictions are all wrong, 1 for one whose predictions are all correct, and $\frac{1}{2}$ for random guessing (the triangle below the diagonal). It is **scale-invariant**, measuring how well the predictions are ranked rather than their absolute values, and **classification-threshold-invariant**, measuring the quality of the predictions whatever threshold is chosen.

Example 2.5 (Four classifiers in ROC space)**.** Each classifier is tested on 100 positives and 100 negatives.

	TP	FN	FP	TN	TPR	FPR	PPV	F_1	ACC
A	63	37	28	72	0.63	0.28	0.69	0.66	0.68
B	77	23	77	23	0.77	0.77	0.50	0.61	0.50
C	24	76	88	12	0.24	0.88	0.21	0.23	0.18
C′	76	24	12	88	0.76	0.12	0.86	0.81	0.82

For instance, for A: $\mathrm{PPV} = 63/91$, $F_1 = 126/191$, $\mathrm{ACC} = 135/200$. B lies on the diagonal, so it is no better than chance. C lies below it, i.e. worse than chance; C′ inverts every decision of C, which swaps TP with FN and FP with TN, so $\mathrm{TPR}' = 1 - \mathrm{TPR} = 0.76$ and $\mathrm{FPR}' = 1 - \mathrm{FPR} = 0.12$: the point is mirrored through $(\frac{1}{2}, \frac{1}{2})$ and C′ is the best of the four.

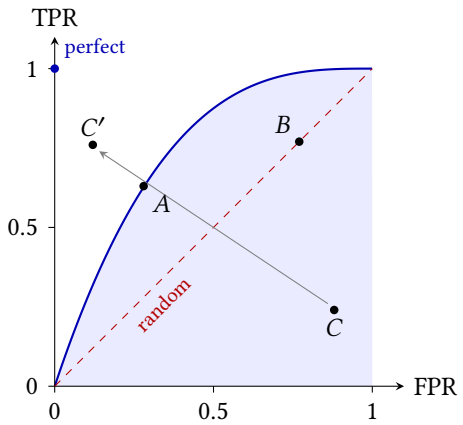


Figure 13: ROC space with the classifiers of example 2.5 and an illustrative ROC curve; the shaded area is its AUC ($= \frac{3}{4}$).

Chest radiographs. A chest X-ray (CXR) shows the lungs and the heart; radiologists compare it with a mental model of a normal CXR, looking e.g. for an enlarged heart (**cardiomegaly**) or for fluid

or air where they should not be. In **CheXpert** a CNN takes the frontal and lateral views and outputs a probability for each of 14 observations (multi-label classification). The labels (positive, negative, uncertain) are extracted automatically from the radiology reports, and the model is compared with radiologists by placing their operating points on the ROC and precision–recall curves of each pathology.

Mammography. X-ray imaging of the breast for the early detection of breast cancer. Radiologists grade lesions with **BI-RADS**, five classes of malignancy risk from 1 (lowest) to 5 (highest) based on shape, density and margin of a lesion, calcifications, distortion and asymmetry, and grade the breast with the **ACR density** classes a (almost entirely fatty) to d (extremely dense); dense breasts lower the sensitivity of mammography. AI tasks are lesion detection, malignant-versus-benign classification and multi-class grading (BI-RADS, ACR). Earlier computer-aided diagnosis (CAD) systems failed in clinical use (no real gain for radiologists, many false positives, poor workflow integration); deep learning reaches human-level performance, but generalisability, workflow integration and interpretability remain open.

Brain age regression. Healthy ageing changes brain volumes following a specific pattern. In Alzheimer's disease (AD) the connections between neurons stop working, which leads to atrophy, i.e. loss of brain volume (fig. 14). The **brain-age gap**, predicted (biological) brain age minus chronological age, is therefore a potential biomarker. A regression model, typically a CNN on whole-brain 2D or 3D MR images, is trained on healthy brains to predict the chronological age, minimising the **mean absolute error** $\mathrm{MAE} = \frac{1}{N}\sum_i |y_i - \hat{y}_i|$. At test time a brain age larger than the chronological age indicates accelerated ageing and points to a possible AD patient.

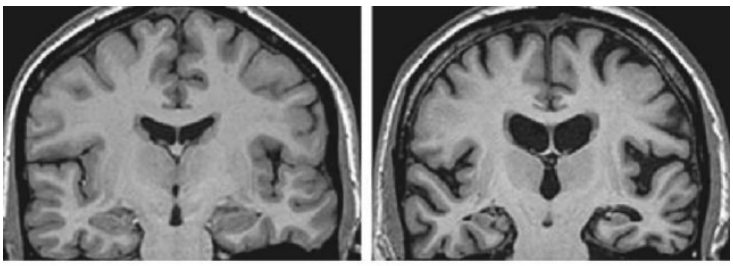


Figure 14: Coronal T1-weighted MR of a healthy and of an Alzheimer's brain: atrophy shows as enlarged ventricles and widened sulci. Source: J. M. Fernández Montenegro, *Alzheimer's Disease Diagnosis Based on Cognitive Methods in Virtual Environments and Emotions Analysis*, 2018.

Bias correction. Age predictions are biased, possibly because of regression dilution: the ages of younger subjects are overestimated and those of older subjects underestimated. The brain-age delta $\Delta = Y_{\text{predicted}} - Y_{\text{chronological}}$ is modelled as a linear function of the chronological age, $\Delta = \alpha\, Y_{\text{chronological}} + \beta$ (slope α, intercept β), and the prediction is corrected by

$$Y_{\text{corrected}} = Y_{\text{predicted}} - (\alpha\, Y_{\text{chronological}} + \beta),$$

so the chronological age enters as a covariate.

A patch-based variant trains one 3D CNN per patch of the brain and combines the P patch predictions y_i in an **ensemble**, by averaging, $y = \frac{1}{P}\sum_{i=1}^{P} y_i$, or by linear regression, $y = w_0 + \sum_{i=1}^{P} w_i y_i$; the per-patch errors also show which brain regions carry most information about age.

3 Advanced Deep Learning Architectures

3.1 Very deep CNNs

The degradation problem. Is a better network just a deeper one? Stacking more plain layers eventually makes things worse: a much deeper plain CNN has a higher *training* error, and hence also a higher test error, than a shallower one, so the problem is not overfitting. A deeper model should not do worse on the training set: copying the layers of a trained shallower model and setting the extra layers to the identity gives a deeper model with the same training error. The deeper solution exists by construction, but the solvers fail to find it, an optimisation difficulty.

ResNet. A **residual block** adds an identity shortcut around a few weight layers (fig. 15). If $H(\boldsymbol{x})$ is the mapping the block should represent, the layers learn the **residual**

$$F(\boldsymbol{x}) = H(\boldsymbol{x}) - \boldsymbol{x}, \qquad \text{so that} \qquad H(\boldsymbol{x}) = F(\boldsymbol{x}) + \boldsymbol{x}.$$

An identity block now only needs $F = 0$, which is easy to reach, so extra layers can no longer hurt the training error. With residual blocks the deeper network is again the better one (a deeper ResNet beats a shallower one, whereas among plain networks the deeper one is worse), and depth can grow to more than a hundred layers.

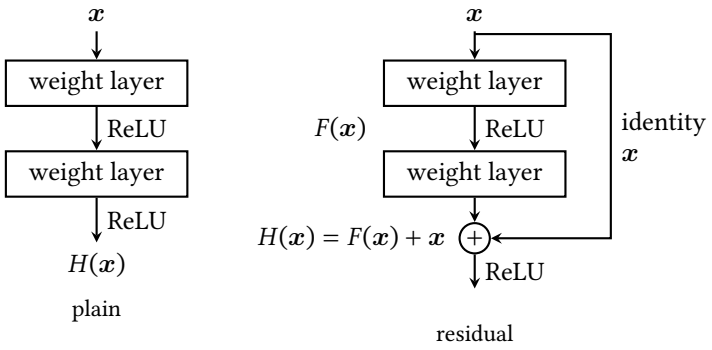


Figure 15: A plain block learns $H(\boldsymbol{x})$ directly; a residual block learns $F(\boldsymbol{x}) = H(\boldsymbol{x}) - \boldsymbol{x}$ and adds the input back.

DenseNet. Inside a **dense block** every layer receives the feature maps of *all* preceding layers, joined by channel-wise concatenation instead of the element-wise addition of ResNet (fig. 16):

$$\boldsymbol{x}_\ell = H_\ell\big([\boldsymbol{x}_0, \boldsymbol{x}_1, \dots, \boldsymbol{x}_{\ell-1}]\big).$$

Each layer adds only k new feature maps, the **growth rate**. Its advantages:

- **parameter and computational efficiency**: a ResNet layer maps C channels to C channels, $O(C \cdot C)$ parameters, whereas layer ℓ of a dense block maps about ℓk channels to k, $O(\ell \cdot k \cdot k)$ parameters, with $k \ll C$;
- **implicit deep supervision**: every layer has a short path to the loss, so the error signal reaches early layers directly (strong gradient flow);
- **more diversified features**: a layer sees all earlier features rather than a sum of correlated ones.

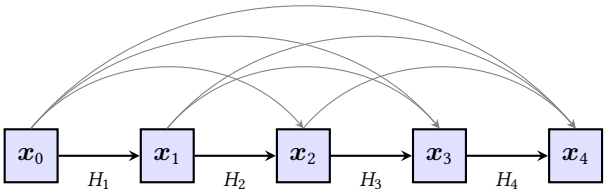


Figure 16: Dense block: each layer H_ℓ gets the concatenation of all earlier feature maps.

CheXNet. A DenseNet-121 (dense blocks separated by convolution and pooling layers, then a linear layer) classifies chest X-rays into 14 pathologies (e.g. pneumonia, cardiomegaly, effusion, pneumothorax, edema), trained on a large labelled dataset. The output is a vector of independent binary labels, one per class c, trained with the sum of binary cross-entropies

$$L(X, y) = \sum_{c=1}^{14} \Big[-y_c \log p(Y_c = 1 \mid X) - (1 - y_c) \log p(Y_c = 0 \mid X) \Big].$$

For pneumonia it was compared with radiologists through the F_1 score (the harmonic mean of precision and recall), and reached a higher F_1 than their average. More generally, modern CNN architectures trade accuracy against size and computation: some are optimised for accuracy, others (e.g. MobileNet, ShuffleNet) for a small number of parameters.

3.2 Recurrent neural networks

Sequences. Feed-forward networks map one input to one output, but much medical data is time-dependent: speech or text (e.g. healthcare records), video (e.g. ultrasound sequences), signals (e.g. ECG). Temporal context matters, and feeding a whole sequence into a feed-forward network at once is memory-inefficient, hard to train and mixes the spatial and temporal dimensions. A **recurrent neural network** (RNN) instead has a loop that lets information persist from one time step to the next, a memory used in particular in language models.

Definition 3.1 (RNN)**.** An RNN has an input, a recurrent hidden and an output layer. It reads a sequence $\dots, \boldsymbol{x}_{t-1}, \boldsymbol{x}_t, \boldsymbol{x}_{t+1}, \dots$ with $\boldsymbol{x}_t \in \mathbb{R}^N$; the hidden state $\boldsymbol{h}_t \in \mathbb{R}^M$ (the memory) and the output $\boldsymbol{y}_t \in \mathbb{R}^P$ are

$$\boldsymbol{h}_t = f_H\big(\mathbf{W}_{IH}\boldsymbol{x}_t + \mathbf{W}_{HH}\boldsymbol{h}_{t-1} + \boldsymbol{b}_h\big), \qquad \boldsymbol{y}_t = f_O\big(\mathbf{W}_{HO}\boldsymbol{h}_t + \boldsymbol{b}_o\big),$$

with input-to-hidden weights $\mathbf{W}_{IH}$, recurrent weights $\mathbf{W}_{HH}$ (also $\mathbf{W}_{\text{rec}}$), hidden-to-output weights $\mathbf{W}_{HO}$, biases $\boldsymbol{b}_h, \boldsymbol{b}_o$, hidden activation f_H (usually tanh) and output activation f_O (usually the sigmoid σ).

Unrolling the loop gives one copy of the unit per time step, all with the same weights, starting from an initial state $\boldsymbol{h}_0$ (fig. 17); stacking such units gives a **deep RNN**.

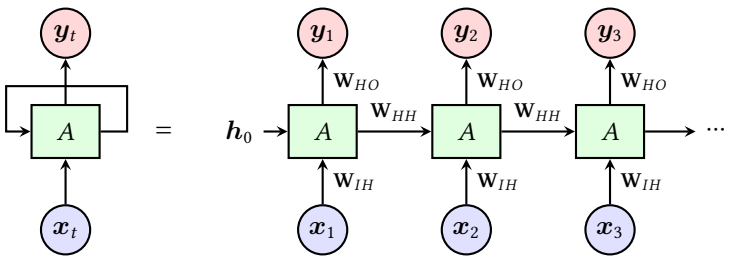


Figure 17: An RNN unit and its unrolled form: the same weights at every time step.

Backpropagation through time. For a sequence $\boldsymbol{x}_1, \dots, \boldsymbol{x}_T$ the forward pass computes $\boldsymbol{h}_t$ and $\hat{\boldsymbol{y}}_t$ for $t = 1, \dots, T$, and the loss is

$$L(\hat{\boldsymbol{y}}, \boldsymbol{y}) = \sum_{t=1}^{T} L(\hat{\boldsymbol{y}}_t, \boldsymbol{y}_t).$$

Its gradient with respect to $\Theta = (\mathbf{W}_{IH}, \mathbf{W}_{HH}, \mathbf{W}_{HO}, \boldsymbol{b}_h, \boldsymbol{b}_o, \boldsymbol{h}_0)$ is computed backwards from $t = T$ to $t = 1$, and the weights are updated by $\Theta \leftarrow \Theta - \eta \nabla\Theta$. Both passes run sequentially through the whole sequence, so computation is slow for long sequences.

Vanishing and exploding gradients. By the chain rule the gradient contains long products in which the recurrent weights appear over and over: with $\boldsymbol{o}_t = \mathbf{W}_{IH}\boldsymbol{x}_t + \mathbf{W}_{HH}\boldsymbol{h}_{t-1} + \boldsymbol{b}_h$,

$$\frac{\partial \boldsymbol{h}_T}{\partial \boldsymbol{h}_1} = \prod_{t=2}^{T} \frac{\partial \boldsymbol{h}_t}{\partial \boldsymbol{h}_{t-1}} = \prod_{t=2}^{T} \operatorname{diag}\left(f_H'(\boldsymbol{o}_t)\right) \mathbf{W}_{HH}.$$

Multiplying many factors smaller than 1 gives ever smaller numbers (**vanishing gradients**, $\mathbf{W}_{HH}$ small); factors larger than 1 give ever larger ones (**exploding gradients**, $\mathbf{W}_{HH}$ large). Remedies:

- against exploding gradients: **truncated backpropagation** (stop after a certain number of steps, so some weights are never updated), penalising or artificially reducing the gradient, **gradient clipping** (a maximum limit on the gradient);
- against vanishing gradients: a suitable weight initialisation, echo state networks, and LSTMs.

A related problem is that the last hidden state hardly depends on the first tokens of a long sequence: information from long ago is difficult to access. Besides LSTMs and Transformers, a **bidirectional RNN** helps: two hidden layers running in opposite directions feed the same output, which thus receives information from both past and future states (fig. 18), at the cost of a more complex backpropagation.

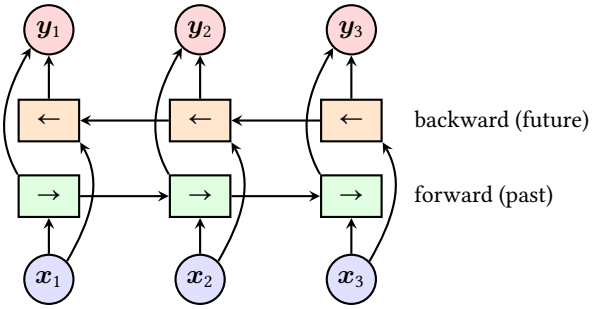


Figure 18: Bidirectional RNN: each output combines a forward and a backward hidden state.

LSTM. A **long short-term memory** unit adds a **cell state** $\boldsymbol{c}_t$, a pipeline flowing through time that is changed only by element-wise operations (fig. 19); put simply, along this path the recurrent weight is 1, so gradients neither vanish nor explode. Each unit has three inputs $(\boldsymbol{x}_t, \boldsymbol{h}_{t-1}, \boldsymbol{c}_{t-1})$ and two outputs $(\boldsymbol{h}_t, \boldsymbol{c}_t)$, and three **gates** (sigmoid valves with values in $(0, 1)$, each with its own weights acting on $[\boldsymbol{h}_{t-1}, \boldsymbol{x}_t]$) control writing to and reading from the memory:

$$\boldsymbol{c}_t = \boldsymbol{f}_t \odot \boldsymbol{c}_{t-1} + \boldsymbol{i}_t \odot \boldsymbol{g}_t, \qquad \boldsymbol{h}_t = \boldsymbol{o}_t \odot \tanh(\boldsymbol{c}_t),$$

where the **forget gate** $\boldsymbol{f}_t$ decides what to keep of the old memory, the **input gate** $\boldsymbol{i}_t$ how much of the candidate $\boldsymbol{g}_t$ (a tanh layer) to write, and the **output gate** $\boldsymbol{o}_t$ what to expose as hidden state; $\odot$ is the element-wise product. Variations add **peephole connections** (the gates also see the cell state), couple forget and input gate (write $1 - \boldsymbol{f}_t$ instead of a separate $\boldsymbol{i}_t$), or drop the cell state altogether: the **gated recurrent unit** (GRU) keeps only the hidden-state pipeline.

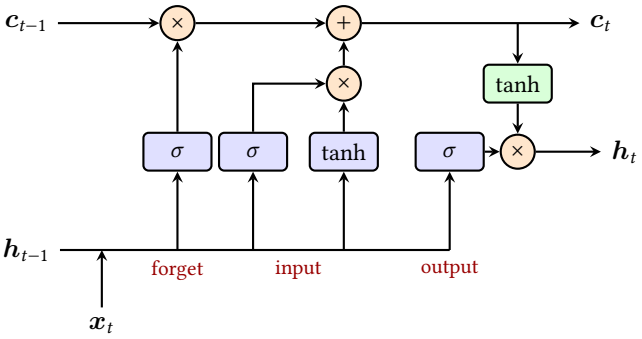


Figure 19: LSTM cell: forget, input and output gates acting on the cell state $\boldsymbol{c}$.

3.3 Transformers

Architecture. The **Transformer** solves most problems of RNNs. It consists of an **encoder** and a **decoder**, each a stack of N identical layers, whose output passes through a linear layer and a softmax (fig. 20); the encoder output is an input of the decoder. Its ingredients are the input and output embeddings, the positional encoding, multi-head attention, feed-forward layers and "Add & Norm" (a residual connection followed by layer normalisation).

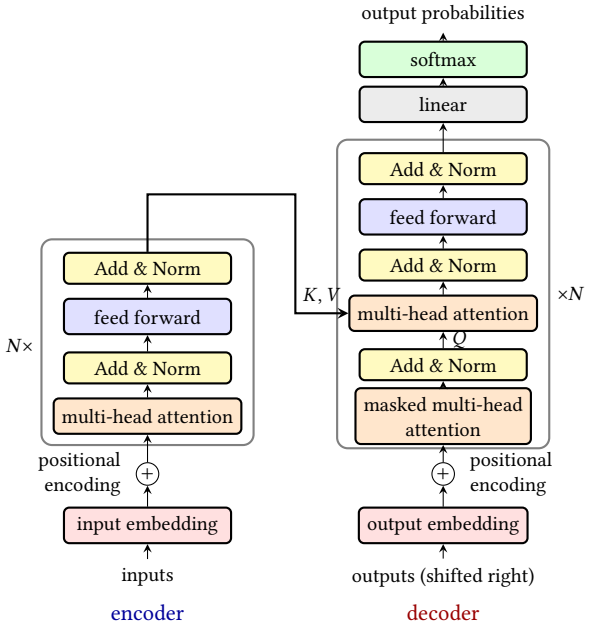


Figure 20: The Transformer: encoder (left) and decoder (right); the encoder output gives keys and values to the decoder's cross-attention.

Embedding and positional encoding. A sentence is split into tokens (words), each token is mapped to its fixed ID in a vocabulary, and the ID to an **embedding** vector of size d_{model}, whose values are learned. Attention by itself ignores word order, so each token also carries its position (words close in the sentence should be treated as close, distant ones as distant): a **positional encoding** of the same size is added to the embedding,

$$\text{PE}(\text{pos}, 2i) = \sin \frac{\text{pos}}{10000^{2i/d_{\text{model}}}}, \qquad \text{PE}(\text{pos}, 2i+1) = \cos \frac{\text{pos}}{10000^{2i/d_{\text{model}}}},$$

for position pos and dimension index i. It is not learned: it is computed once and reused for every sentence, in training and inference. Sine and cosine give a continuous, regular pattern from which relative positions are easy to recognise.

Definition 3.2 (Self-attention)**.** Stack the n token vectors of a sequence as rows of matrices Q (queries), K (keys) and V (values) of size $n \times d_k$. Then

$$\text{Attention}(Q, K, V) = \text{softmax}\Big(\frac{QK^\top}{\sqrt{d_k}}\Big)V,$$

where the softmax acts on each row. The $n \times n$ matrix of attention weights has rows summing to 1, and row j of the $n \times d_k$ result is a weighted mean of the values (fig. 21), which encodes the meaning of token j (embedding), its position (positional encoding) and its interaction with the other tokens.

The names come from a database lookup: a query is compared with all keys, and the values of the best-matching keys are returned, here softly weighted by the scaled dot products. In the simplest form $Q = K = V$ is the input matrix itself, so self-attention relates every word to every other word and

- is **permutation equivariant**: permuting the input tokens permutes the output rows in the same way, leaving each row unchanged;
- has no parameters: the interactions are driven by the embedding and positional encoding (this changes in multi-head attention);
- gives the largest weights typically on the diagonal (each word's dot product with itself). Entries can be manipulated before the softmax: an entry set to $-\infty$ gets weight $e^{-\infty} = 0$.

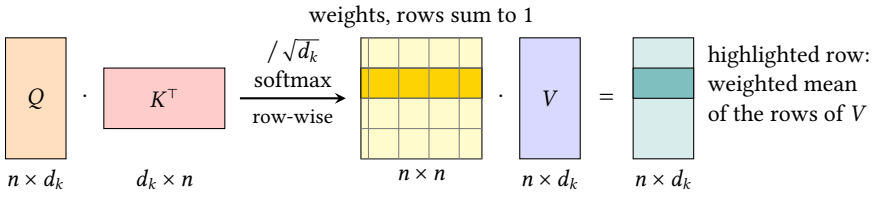


Figure 21: Self-attention: the softmax of the scaled scores weights the rows of V.

Multi-head attention. With h heads and $d_k = d_v = d_{\text{model}}/h$, each head attends in its own learned projection, and the heads are concatenated and projected back:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)\, W^O, \qquad \text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V).$$

The learned matrices W_i^Q, W_i^K, W_i^V, W^O let different heads capture different relations between words (fig. 22).

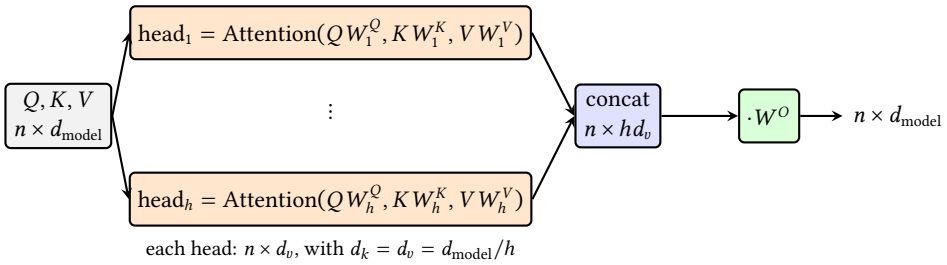


Figure 22: Multi-head attention: h heads in parallel, concatenated and projected by W^O.

Layer normalisation. "Add & Norm" adds the sublayer's input (residual connection) and normalises each item j separately with its own mean μ_j and variance σ_j^2 over its features,

$$\hat{x}_j = \frac{x_j - \mu_j}{\sqrt{\sigma_j^2 + \epsilon}},$$

then rescales with learnable parameters γ (multiplicative) and β (additive), since values confined to a fixed range may be too restrictive. Unlike batch normalisation, the statistics are computed per item, not per feature across the batch.

Decoder. The lower part (embedding, positional encoding) is as in the encoder. Its first attention is **masked multi-head attention**: the model is causal, the output at a position may depend only on earlier positions and must not see future words, so all entries above the diagonal of the attention matrix are set to $-\infty$ before the softmax (their weights become 0, fig. 23). The second attention is **cross-attention**: keys and values come from the encoder output, the queries from the masked attention. A final linear layer maps d_{model} to the vocabulary size (the outputs are the **logits**), and a softmax gives the probability of each vocabulary token.

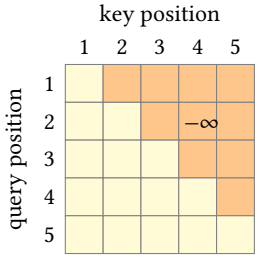


Figure 23: Causal mask: query position j may attend only to key positions $1, \dots, j$.

Training and inference. In training everything happens in one time step: the decoder input is the target sentence preceded by a start token ⟨SOS⟩ ("shifted right"), and the softmax output at every position is compared by a cross-entropy loss with the target followed by ⟨EOS⟩; the mask prevents each position from seeing its own answer. At inference the encoder runs once; the decoder starts from ⟨SOS⟩ and at each step takes the most probable token at the last position, appends it to its input and repeats until it emits ⟨EOS⟩.

Example 3.3 (Translation)**.** Input "⟨SOS⟩ I love you very much ⟨EOS⟩". Training: the decoder reads "⟨SOS⟩ Ti amo molto" and must output "Ti amo molto ⟨EOS⟩" at once. Inference: from "⟨SOS⟩" it outputs "Ti"; from "⟨SOS⟩ Ti" it outputs "amo"; then "molto"; then ⟨EOS⟩, and it stops. The encoder output computed in the first step is reused in all later steps.

Language models. Transformers learn sequence-to-sequence maps (they were first used for machine translation) and are the backbone of large language models (LLMs) such as BERT, ChatGPT, Claude and Llama. Medical NLP tasks fall into two families. **Discriminative tasks** turn text (questions, clinical notes, papers, health records) into structured information, e.g. question answering, or extracting symptoms, drugs and diagnoses and linking them to a terminology such as ICD codes. **Generative tasks** produce new text, e.g. summarising the findings of a radiology report into its impression, or simplifying medical text for readability.

Vision Transformer (ViT). Earlier, attention in vision was used together with CNNs or to replace some of their components. ViT removes convolutions entirely and applies a pure Transformer encoder to image patches treated like tokens (fig. 24):

1. split the image into patches (e.g. 16×16 pixels) and flatten each one;
2. project each patch linearly to dimension D;
3. prepend a learnable `[class]` embedding and add learned position embeddings;
4. pass the sequence through L Transformer encoder layers;
5. an MLP head reads the output at the `[class]` position and predicts the class.

It is trained on image classification in a supervised way, and reached excellent results compared with the best CNNs at lower training cost.

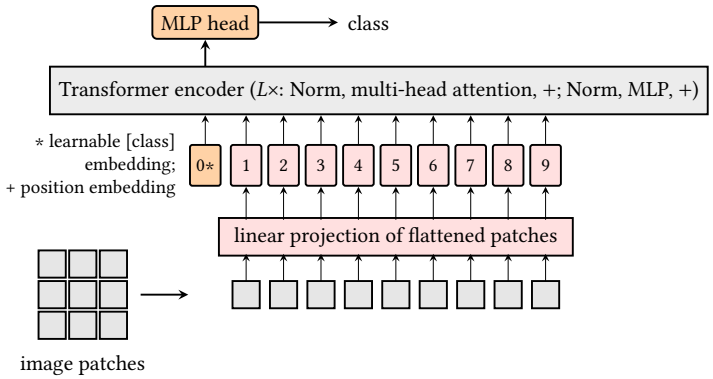


Figure 24: Vision Transformer: patches become tokens; the MLP head classifies from the `[class]` token.

Pre-training and fine-tuning. ViT is typically pre-trained on large datasets (also self-supervised) and fine-tuned on smaller downstream tasks: the pre-trained prediction head is replaced by a zero-initialised $D \times K$ layer for K downstream classes. Fine-tuning at a higher resolution than pre-training often helps; the patch size stays the same, so the sequence gets longer. The Transformer handles any sequence length, but the pre-trained position embeddings then lose their meaning, which is fixed by 2D interpolation according to their location in the original image. This resolution adjustment and the patch extraction are the only places where an inductive bias about the 2D structure of images is put in by hand.

Inspecting a trained ViT shows that it attends to the image regions semantically relevant for the class, and that the learned position embeddings of patches in the same row or column are similar, i.e. the model learns the 2D layout.

Swin Transformer. ViT keeps the same coarse resolution (patches of 16× the pixel size) in every layer. The **Swin Transformer** uses a hierarchical, CNN-like pyramid that merges patches and so downsamples the feature maps gradually (4×, 8×, 16×, …), which helps dense downstream tasks such as object detection and semantic segmentation (fig. 25). Self-attention is computed only within local windows (W-MSA), and successive blocks shift the windows (SW-MSA) so that information flows between neighbouring windows.

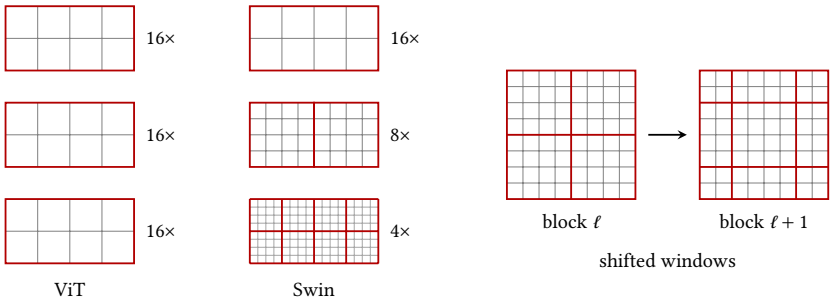


Figure 25: ViT keeps 16× patches throughout; Swin merges patches (4× to 16×) and attends within local (red) windows, shifted in the next block.

Case study: COVID-19 on chest X-rays. A multi-task ViT for diagnosis and severity quantification, made of

- a **shared part**: a pre-trained CNN backbone (a DenseNet) extracts low-level chest X-ray features, which become the tokens of a Transformer encoder with a `[class]` token;
- two **task-specific heads**: a classification head on the `[class]` output (normal, other infection, COVID-19), and a map head whose severity map, restricted to the lungs by a mask, is max-pooled over lung regions into severity scores trained against the radiologists' region-wise labels.

4 Image Segmentation I

4.1 Segmentation and its challenges

Image analysis tasks. Tasks differ in their output. **Image classification** assigns a category to the whole image ("bird"), **object recognition** a finer category to an object ("kingfisher"); **object detection** returns coordinates of each object, e.g. its centroid, **object localisation** a bounding box around it; **image captioning** returns a text. **Semantic segmentation** returns a **label map**: every pixel gets a label, and each segmented region carries a semantic meaning (liver, left kidney, background) instead of being just a cluster of coherent pixels. With two labels it is a foreground/background segmentation, with more a multi-class one.

In medicine segmentation is used to compute quantitative **biomarkers** (e.g. the volume of the ventricular cavity), to determine the precise location and extent of an organ or a tissue such as a tumour (e.g. for radiation therapy), and to build 3D models for simulation (e.g. of an abdominal aortic aneurysm, to simulate the stress and strain in its wall).

Challenges. Several properties of medical images hamper segmentation algorithms:

- **noise**, strong for instance in ultrasound (where it is a separate effect from speckle, attenuation and reflection);
- the **partial volume effect**: the scene is sampled coarsely and each pixel collects signal from a finite area, so a pixel on the boundary of an object mixes both tissues and takes an intermediate intensity (fig. 26). More generally the image is the true object convolved with the **point spread function** (PSF), the response of the imaging system to a point source, which blurs edges and lowers the peak intensity of small objects;
- **intensity inhomogeneities**: the same tissue has different intensities in different parts of the image, typically in MRI (inhomogeneous magnetic field) and ultrasound (attenuation);
- **anisotropic resolution**: a 3D image is made of **voxels** (volume elements), the 3D analogue of pixels: a pixel has coordinates (x, y), a voxel (x, y, z), and each is the patch or box of tissue represented by one intensity value. In most 3D modalities (CT, MRI) the distance between slices is much larger than the resolution within a slice, so voxels are elongated boxes (**anisotropic**) instead of cubes (**isotropic**), and views across the slices look blocky;
- **imaging artefacts**, e.g. streaks in CT caused by metal implants such as dental fillings, or susceptibility artefacts in MR caused by metallic particles;
- **limited contrast**: different tissues may have similar physical properties and hence similar intensities (e.g. a thrombus in the wall of an aortic aneurysm in CT angiography), so purely intensity-based algorithms fail or **leak** into the adjacent tissue (fig. 29b);
- **morphological variability**: shapes differ strongly between patients, which makes it hard to build in prior information or shape models.

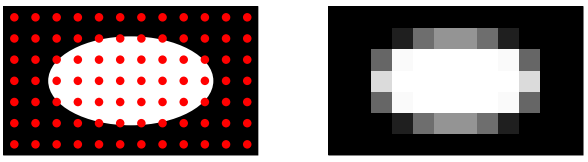


Figure 26: Partial volume effect: an object and its pixel image, where boundary pixels are mixtures.

4.2 Evaluation

Ground truth and gold standard. A **ground truth** is the true reference a method is compared with, e.g. the true segmentation boundary (or, in registration, the optimal transformation). It is not available in practice, so one is made up: with **synthetic phantoms** (simulated images) or **physical phantoms**, objects of known geometry that are scanned, such as a gel mould of a brain imaged with

MR, CT and ultrasound. On patient images one uses a **gold standard**, the manual segmentation by an expert such as an experienced clinician. It requires training, is tedious and time-consuming, and suffers from **intra-observer variability** (the same observer disagrees with themselves on different occasions) and **inter-observer variability** (different observers disagree). Remedies are to let one observer segment repeatedly and several experts segment the same image, and to quantify their agreement.

Pixel-wise measures. Precision, accuracy and robustness, and the classification metrics of section 2.3, carry over by treating every pixel as one binary decision: compared with the gold standard, a pixel is TP if both segmentations mark it as object, FP if only the automatic one does (e.g. where it leaks), FN if only the reference does, TN otherwise, and P is the number of object pixels of the reference. ACC, PPV, TPR, TNR and F_1 (definition 2.4) are then computed from these counts.

Overlap. Let A and B be the *regions* labelled as object by the two segmentations (reference and prediction), and $|\cdot|$ their number of pixels (area or volume).

Definition 4.1 (Jaccard index and Dice coefficient)**.**

$$\mathrm{JI} = \frac{|A \cap B|}{|A \cup B|}, \qquad \mathrm{DSC} = \frac{|A \cap B|}{(|A| + |B|)/2} = \frac{2|A \cap B|}{|A| + |B|}.$$

The **Jaccard index** is also called Jaccard similarity coefficient, Tanimoto coefficient or **intersection over union** (IoU); the **Dice coefficient** is also called Sørensen index, **Dice similarity coefficient** (DSC) or intersection over mean volume.

Both lie in $[0, 1]$ (fig. 27a): they are 0 for no overlap ($A \cap B = \emptyset$) and 1 for perfect overlap ($A = B$, where $|A \cap B| = |A \cup B| = (|A| + |B|)/2$). With A the prediction and B the reference, $|A \cap B| = \mathrm{TP}$, $|A| = \mathrm{TP} + \mathrm{FP}$ and $|B| = \mathrm{TP} + \mathrm{FN}$, so the Dice coefficient is the F_1 score:

$$\mathrm{DSC} = \frac{2\,\mathrm{TP}}{2\,\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} = F_1.$$

Overlap measures say how much the regions overlap, not how their boundaries differ, and they are sensitive to the overall size and shape of the volume: a translated copy of a reference disc and differently deformed versions of it can all have the identical DSC.

Surface distances. Here A and B are *surfaces*, the boundaries (contours in 2D) of the two segmentations, seen as sets of points.

Definition 4.2 (Hausdorff and average surface distance)**.** With the directed distances

$$h(A, B) = \max_{a \in A} \min_{b \in B} \|a - b\|, \qquad d(A, B) = \frac{1}{|A|} \sum_{a \in A} \min_{b \in B} \|a - b\|,$$

the **Hausdorff distance** and the (symmetric) **average surface distance** are

$$\mathrm{HD} = \max\big(h(A, B), h(B, A)\big), \qquad \mathrm{ASD} = \frac{d(A, B) + d(B, A)}{2}.$$

$\min_{b \in B} \|a - b\|$ is the distance from a to the nearest point of B; $h(A, B)$ takes the worst point of A and $d(A, B)$ the mean over A. Neither is symmetric (a protrusion of B far from A does not change $h(A, B)$), hence the maximum and the mean over both directions. HD is the worst-case boundary error, determined by a single farthest point; ASD is the typical one (fig. 27b).

Both are computed with **Euclidean distance maps**: the distance map of B assigns to each pixel its distance to the nearest point of B, so B is its zero level set. Reading this map at the boundary pixels of A gives $\min_{b \in B} \|a - b\|$ for every $a \in A$; their mean is $d(A, B)$ and their maximum $h(A, B)$. Swapping the roles, with the distance map of A, gives $d(B, A)$ and $h(B, A)$.

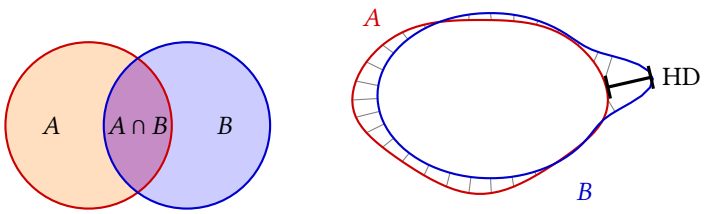


Figure 27: (a) Regions A, B and their overlap. (b) Contours A, B: distances from A to B and the Hausdorff distance.

4.3 Segmentation methods

Segmentation methods fall into families: **intensity-based** (e.g. thresholding), **region-based** (e.g. region growing), **graph-based** (e.g. graph cuts), **active contours** (e.g. level sets), **atlas-based** (e.g. multi-atlas label propagation) and **learning-based** (e.g. random forests, convolutional neural networks). Below, $I(x)$ is the intensity of pixel x.

Thresholding. **Simple thresholding** labels as object every pixel with $I(x) \geq T$; **upper-lower (UL) thresholding** every pixel with $T_l \leq I(x) \leq T_u$. The thresholds are chosen on the intensity histogram, where different tissues form different peaks (fig. 28). For example, on a head CT with a haemorrhage a low threshold separates the head from the surrounding air, a high one keeps only the skull, and a window around the intensities of the haemorrhage selects it, but also scattered brain pixels whose intensities fall into the same window.

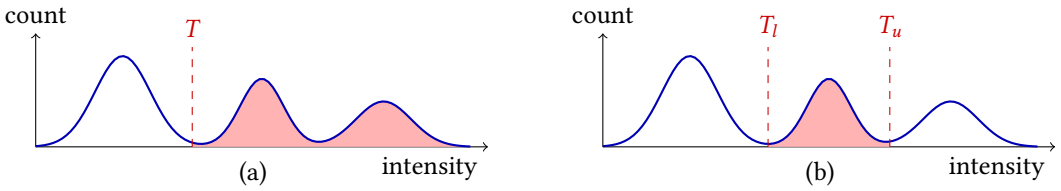


Figure 28: Histogram with (a) a simple threshold and (b) lower and upper thresholds; shaded intensities are labelled object.

Region growing. Start from one or more **seed points**, usually selected by the user, and grow a region by adding neighbours whose intensity satisfies a threshold criterion, until no neighbour can be added (algorithm 1).

Algorithm 1 Region growing

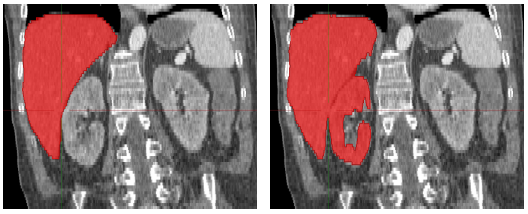
Require: image I, seed points S, intensity criterion (e.g. $T_l \leq I(x) \leq T_u$)
Ensure: region R

1: $R \leftarrow S$
2: **while** some neighbour $x \notin R$ of a pixel of R satisfies the criterion **do**
3: $\quad R \leftarrow R \cup \{x\}$
4: **end while**

The result is a connected region containing the seed, and pixels that pass the criterion but are not connected to it are left out (fig. 29a). The result depends on the seed: on the head CT above, a seed in the haemorrhage yields the haemorrhage, a seed in the brain tissue the whole brain. Where contrast is low the region can leak into a neighbouring organ (fig. 29b).

12	14	13	40	42	15
13	45	47	44	41	14
11	46	52	48	16	12
14	43	49	15	13	44
12	15	47	14	12	46
13	12	14	13	11	15

(a)



(b)

Figure 29: Region growing. (a) Red seed, interval $[35, 55]$: grown region orange, unreached pixels in the interval grey. (b) Liver in CT: manual segmentation and leakage into the kidney.

Graph cuts. Segmentation is formulated as a **minimum cut** in a graph, computed with max-flow/min-cut algorithms (fig. 30). Every pixel is a node; **n-links** connect neighbouring pixels, and two extra terminal nodes, the source s (object) and the sink t (background), are connected to every pixel by **t-links**. Each edge has a cost, and a **cut** is a set of edges whose removal separates s from t: afterwards every pixel is still connected to exactly one terminal, which gives its label. The minimum cut is the cut of least total cost. Labelling pixel i as background or object cuts its t-link to the other terminal, at the cost

$$\psi_i(0) = -\ln \mathbb{P}(y_i \mid \text{Bkg}), \qquad \psi_i(1) = -\ln \mathbb{P}(y_i \mid \text{Obj}),$$

where y_i is the intensity of pixel i: a label is cheap where the intensity is likely under that class. The n-links are cut wherever the boundary passes between two neighbours, so the boundary is drawn where the cut is cheapest, i.e. between dissimilar neighbouring pixels, as at the edge between an object and its background. In interactive use the input are **seed brushes**, strokes by which the user marks some object and some background pixels and from which the intensity distributions $\mathbb{P}(y \mid \text{Obj})$ and $\mathbb{P}(y \mid \text{Bkg})$ are obtained; the cut then delineates e.g. a brain haemorrhage. **Multi-label graph cuts** segment several structures at once (liver, kidneys, spleen, background, one brush each), and the user corrects errors interactively by adding brushes and recomputing.

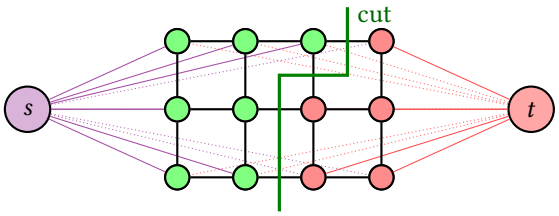


Figure 30: Graph cut on a pixel grid: n-links, t-links to s and t, and the minimum cut separating object from background.

Decision trees. A classification tree sends a feature vector $\boldsymbol{v}$ from the root to a leaf: each internal node applies a binary test to $\boldsymbol{v}$ and passes it to the left or right child, and each leaf stores a class distribution $p(c \mid \boldsymbol{v})$, the class frequencies of the training points that reached it. The tests split the feature space into regions, one per leaf (fig. 31). Training is a **greedy, randomised optimisation**, run recursively from the root on the whole training set (algorithm 2): each node chooses its test among a set of random candidates (e.g. random lines in the feature plane). A **random forest** is an ensemble of T such trees, trained with different randomness, whose output averages the trees:

$$p(c \mid \boldsymbol{v}) = \frac{1}{T} \sum_{t=1}^{T} p_t(c \mid \boldsymbol{v}).$$

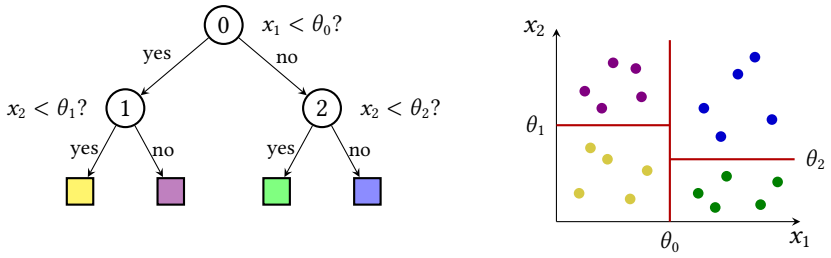


Figure 31: A decision tree with axis-parallel tests and the partition of the feature plane into its leaves.

Algorithm 2 Training a decision tree, called as TrainNode(training set)

```
1:  function TrainNode(S)                          ▷ S: training points reaching the node
2:      if a stopping criterion is met then
3:          make the node a leaf storing the class frequencies of S as p(c | v)
4:      else
5:          draw a set of random candidate tests
6:          keep the test that best separates the classes of S
7:          split S by this test into S_L and S_R
8:          TrainNode(S_L); TrainNode(S_R)          ▷ left and right child
9:      end if
10: end function
```

Brain tumour segmentation with random forests. Each voxel of a multi-channel 3D MR image (T1, contrast-enhanced T1, T2, FLAIR and two diffusion channels) is classified as active tumour cells, necrotic core, edema or background. The tests use **spatial context**: a test compares two boxes at offsets from the voxel, possibly in different channels, $\phi(\text{box}_1) - \phi(\text{box}_2) > \theta$, where ϕ is a feature of the box such as its mean intensity. In **training** the test of each node is selected by optimising over randomised features (offsets, box sizes, channels, threshold θ) on manually segmented images; in **testing** each voxel traverses the learned trees, and the averaged leaf distributions give its class. The results are close to the manual segmentations.

Comparison. The methods trade automation against robustness:

Method	Idea	Advantages	Disadvantages
Thresholding	label pixels by intensity range	simple, fast	regions must be homogeneous and distinct; consistent thresholds across images are hard to find; leakage, isolated pixels, rough boundaries
Region growing	grow a connected region from seeds by an intensity criterion	relatively fast; connected region	regions must be homogeneous; leakage, rough boundaries; needs user seeds
Graph cuts	minimum s–t cut of the pixel graph	accurate; reasonably efficient, interactive	semi-automatic (needs user input); tuning parameters hard to select
Random forests	classify each voxel with an ensemble of trees on context features	robust and accurate (ensemble); efficient; fully automatic	shallow model, no hierarchical features; no guarantee of connectedness

4.4 Unsupervised segmentation

The goal is to segment brain MR images into soft tissues, white matter, grey matter and cerebrospinal fluid (CSF), without labelled training data, by **clustering** the voxel intensities.

K-Means. The algorithm alternates between assigning points to their closest centroid and moving each centroid to the mean of its points (algorithm 3).

Algorithm 3 K-Means

Require: points $x_1, \dots, x_N$, number of clusters K
Ensure: centroids $\mu_1, \dots, \mu_K$ and/or assignments $c_1, \dots, c_N$
initialise $\mu_1, \dots, \mu_K$, e.g. as K distinct input points chosen at random
repeat
  for $n = 1, \dots, N$ **do** ▷ assignment step
    $c_n \leftarrow$ index $k \in \{1, \dots, K\}$ of the centroid closest to x_n
  end for
  for $k = 1, \dots, K$ **do** ▷ update step
    $\mu_k \leftarrow$ average of the points with $c_n = k$
  end for
until a stopping criterion is met, e.g. the assignments no longer change

Definition 4.3 (K-Means cost)**.** With $z_{nk} = 1$ if point n is assigned to cluster k and 0 otherwise, K-Means solves

$$\min_{\{z_{nk}\},\{\mu_k\}} J = \sum_k \sum_n z_{nk} \|x_n - \mu_k\|^2 \qquad \text{s.t.} \quad z_{nk} \in \{0, 1\}, \quad \sum_k z_{nk} = 1 \ \ \forall n.$$

The two steps minimise J alternately. For fixed centroids, point n contributes $\sum_k z_{nk} \|x_n - \mu_k\|^2$, which is smallest with $z_{nk} = 1$ for its closest centroid (assignment step). For fixed assignments, J is a convex quadratic in μ_k and

$$\frac{\partial J}{\partial \mu_k} = -2 \sum_n z_{nk}(x_n - \mu_k) = 0 \iff \mu_k = \frac{\sum_n z_{nk} x_n}{\sum_n z_{nk}},$$

the mean of the points of cluster k (update step). So J never increases, but the algorithm can stop in a poor local minimum: a bad initialisation can yield a bad clustering. The remedy is to rerun it from multiple random initialisations and keep the result with the lowest cost. The cost also explains its caveats: every point goes to the nearest centroid whatever the spread of the clusters, so K-Means prefers clusters of roughly the same size and may cut through an intuitive low-density gap (fig. 32).

Gaussian mixture models. Instead, each cluster is represented by a statistical model, e.g. a (multivariate) Gaussian, and the data by a mixture of them. For intensities, the **Gaussian mixture model** (GMM) is the weighted sum of normal densities

$$p(x) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(x \mid \mu_k, \sigma_k^2),$$

with mixing weights $\pi_k \geq 0$, $\sum_k \pi_k = 1$, so that p is a density. Each cluster has its own spread and weight, so a GMM fitted by EM can separate a wide cluster from small tight ones where K-Means fails (fig. 32). On a brain MR histogram with $K = 3$, K-Means and the GMM give similar tissue maps; with more components the histogram is fitted more closely, but the segmentations break into many classes without anatomical meaning.

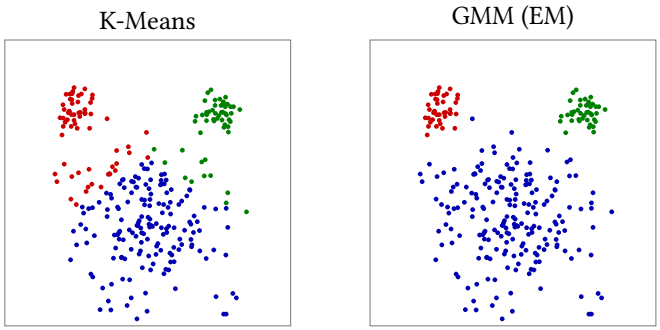


Figure 32: K-Means and a GMM with $K = 3$ on the same illustrative data (colours: clusters).

Expectation-maximisation. **EM** is an iterative method to estimate the parameters of a statistical model that depends on unobserved **latent variables**. For the GMM the data are $X = \{x_n\}_{n=1}^{N}$, the parameters $\theta = \{\pi_k, \mu_k, \sigma_k\}$ and the latent variables $Z = \{z_{nk}\}$, with $z_{nk} = 1$ iff point n belongs to cluster k, the same indicators as in K-Means. EM alternates between the **E-step**, which determines $p(Z \mid X, \theta)$, i.e. soft assignments of the points to the clusters, and the **M-step**, which updates θ. It greedily maximises a lower bound of the (marginal) log-likelihood $\mathcal{L}(\theta) = \log p(X \mid \theta)$.

Number of clusters. K is chosen either by the **elbow method**, plotting the cost against the number of clusters and taking the point after which adding clusters hardly lowers it, or by an **application-based rationale**: the intensity histogram of a T1-weighted brain MR image has three peaks, CSF (darkest), grey matter and white matter (brightest), so $K = 3$.

5 Image Segmentation II

5.1 From dense classification to U-Net and nnU-Net

Patch classification and fully convolutional segmentation. A segmentation can be obtained by classifying every pixel or voxel from a surrounding image patch. A CNN extracts features of the patch and a classifier predicts the label of its centre; sliding the patch across the image produces a label map. Neighbouring patches overlap heavily, so this repeats almost the same convolutions many times. A **fully convolutional network** (FCN, fig. 11) instead shares those computations over the whole image and returns a spatial map of class scores in one forward pass. It is trained end-to-end against the reference label map.

Context and localisation. Repeated pooling gives deep features a large receptive field and hence the semantic context needed to recognise an organ, but loses the fine spatial information needed to delineate its boundary. Shallow, high-resolution features retain edges but see too little context to distinguish the target reliably. An **encoder–decoder** first contracts the spatial representation to extract context, then upsamples it to predict at image resolution, using learned transposed convolutions (fig. 9). In the original FCNs, predictions from coarse and finer layers are fused by addition: FCN-32s upsamples the deepest prediction directly, whereas FCN-16s and FCN-8s progressively incorporate shallower predictions to refine localisation.

U-Net. The **U-Net** combines a contracting encoder and an expanding decoder through **long-range skip connections** (fig. 33). Each encoder stage applies convolutions and downsamples, typically doubling the number of channels; the bottleneck holds the lowest-resolution semantic representation. Each decoder stage upsamples, reduces channels and concatenates the encoder features at the corresponding resolution, then fuses them with convolutions. The final 1×1 convolution maps features to class scores.

In the original architecture, valid convolutions shrink feature maps, so encoder features are cropped to match the decoder before concatenation.

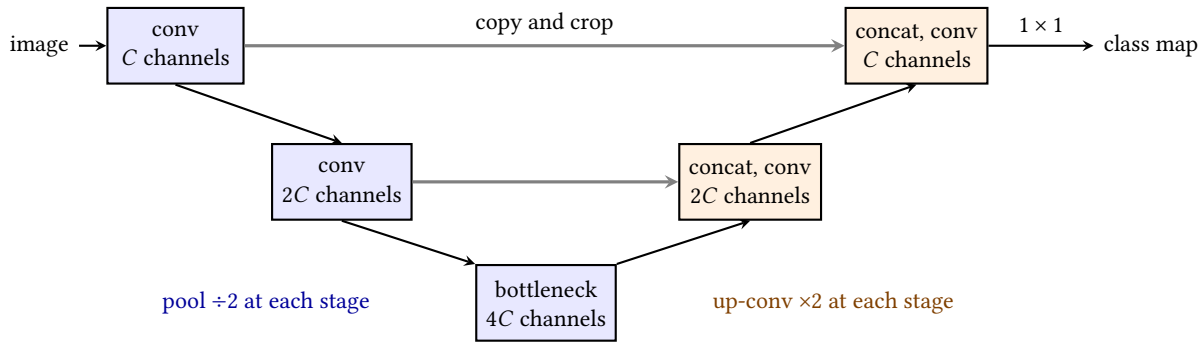


Figure 33: U-Net: blue encoder stages supply spatial detail to orange decoder stages; grey arrows concatenate matching-resolution features. Spatial sizes are schematic.

Concatenation versus addition. Let e be an encoder feature map and d the upsampled decoder map, both of size $H \times W \times C$. Concatenation gives $z = [e, d]$ of size $H \times W \times 2C$. Splitting the next convolution's weights along the input channels shows how the two sources can be weighted independently:

$$y = \sigma(W * z + b) = \sigma(W_e * e + W_d * d + b).$$

Adding first would give $\sigma(W * (e + d) + b)$, with the same subsequent filter acting on both streams. U-Net thus retains both representations until their learned fusion.

nnU-Net. Good segmentation depends on the whole pipeline, including preprocessing, patch size, training and postprocessing. **nnU-Net** configures this pipeline for a new biomedical dataset using its **data fingerprint**: image modality, intensity distribution, voxel spacings and median image shape. Its choices fall into three groups:

- **Blueprint parameters** are fixed across datasets: a U-Net template, instance normalisation, LeakyReLU, the training schedule and optimiser, augmentation, the Dice-plus-cross-entropy loss and **deep supervision** (training losses at intermediate decoder outputs).
- **Inferred parameters** follow heuristic rules based on the fingerprint and GPU memory: resampling and target spacing, intensity normalisation, patch and batch sizes, and network topology (including the number of downsampling stages).
- **Empirical parameters** are selected by cross-validation on training cases: postprocessing and the best configuration or ensemble among 2D, full-resolution 3D and cascaded 3D U-Nets.

For CT, intensities are clipped using dataset percentiles and z-score normalised with global foreground mean and standard deviation from the training set; other modalities use per-image mean and standard deviation. The 3D cascade is activated when the full-resolution patch covers less than 12.5% of the median resampled image volume. It uses a low-resolution segmentation to guide the full-resolution network.

5.2 3D volumetric, multi-scale and attention CNNs

Processing a volume. There are three main compromises between spatial context and computational cost. Kernel sizes below are for one output channel; C_{in} is the number of input channels.

Input	Mechanism	Advantage	Limitation
2D	independent slices; $K \times K \times C_{\text{in}}$ kernels	low memory, larger batches, 2D pre-training	no through-plane context; discontinuous 3D boundaries
2.5D	adjacent slices as channels, or fusion of axial, sagittal and coronal views	local 3D context at near-2D cost	limited through-plane range; asymmetric receptive field
3D	subvolumes with $K \times K \times K \times C_{\text{in}}$ kernels	anatomical context in all directions	feature-map memory grows with DHW; small patches and batches

Full 3D processing uses inter-slice information to reduce discontinuities, but does not by itself enforce connectivity.

Anisotropic resolution. Thick slices mean that a cubic kernel in voxel coordinates covers very different physical distances in-plane and through-plane. Resampling to isotropic spacing allows isotropic kernels, at the cost of more memory and possible through-plane blurring. Alternatively, early layers use $3 \times 3 \times 1$ convolutions and $2 \times 2 \times 1$ pooling, preserving the coarse slice axis until the effective spacings become comparable.

Normalisation with small batches. The memory cost of 3D feature maps often limits training to one or two volumes per batch. Batch normalisation can then have unreliable statistics; instance and group normalisation avoid dependence on the other samples. For a feature tensor of shape $B \times C \times D \times H \times W$, the statistics are computed over:

Normalisation	Axes averaged over	Statistics kept separately for
Batch (BN)	B, D, H, W	each channel
Instance (IN)	D, H, W	each sample and channel
Group (GN), G groups	$C/G, D, H, W$	each sample and channel group

3D U-Net and sparse labels. **3D U-Net** replaces the 2D operations of U-Net with their volumetric counterparts. It can learn from a few annotated orthogonal slices by masking the training loss: the voxel weight $w(x)$ is zero wherever the annotation is missing. Unlabelled voxels therefore contribute no loss, while inference still predicts a dense 3D segmentation.

V-Net and local skip connections. **V-Net** is a volumetric encoder–decoder with residual blocks, learned downsampling through stride-2 convolutions rather than max pooling, and direct optimisation of soft Dice overlap to address class imbalance. Its short-range residual shortcuts improve gradient flow within stages, whereas U-Net's long-range skips transfer spatial detail across the bottleneck. For a chain $x_{i+1} = x_i + F(x_i, W_i)$, the identity contribution is visible in

$$\frac{\partial \mathcal{L}}{\partial x_\ell} = \frac{\partial \mathcal{L}}{\partial x_L} \left(I + \frac{\partial}{\partial x_\ell} \sum_{i=\ell}^{L-1} F(x_i, W_i) \right).$$

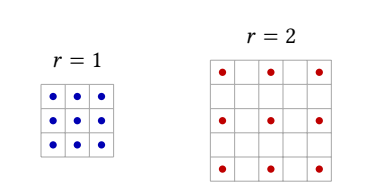
Dense blocks (fig. 16) instead concatenate earlier features within a block, encouraging feature reuse and parameter efficiency when annotated data are scarce. These connections improve optimisation, with an activation-memory cost.

Dilated convolutions. A **dilated** or **atrous convolution** expands the receptive field without additional kernel weights or spatial downsampling. With dilation rate r, adjacent weights are r input positions apart:

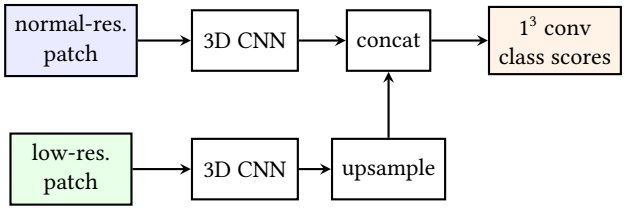
$$(F *_r k)(p) = \sum_{s+rt=p} F(s)k(t), \qquad K_{\text{eff}} = K + (K-1)(r-1).$$

A $3 \times 3 \times 3$ kernel still has 27 spatial weights, but spans a 5^3 receptive field for $r = 2$ and 9^3 for $r = 4$. **Atrous spatial pyramid pooling** (ASPP) applies several dilation rates in parallel and concatenates their outputs to combine scales. Repeated sparse sampling can cause **gridding artifacts**, leaving gaps in the sampled context.

DeepMedic. Another way to enlarge context is to process the image at several resolutions in parallel (fig. 34). **DeepMedic** combines a normal-resolution 3D pathway, retaining local anatomy, with pathways fed downsampled images, seeing a larger physical region. Low-resolution features are upsampled to align with the normal-resolution features, concatenated, then classified with $1 \times 1 \times 1$ convolutions. The extra context improves localisation. Feature maps in the normal-resolution path can respond to anatomical detail, while the coarse path learns broader location information. The same design applies to brain lesions, cardiac structures and placenta segmentation.



(a) Same nine weights, wider support



(b) Same target region, complementary local and broad context

Figure 34: Two ways to enlarge context. (a) A 3×3 kernel samples a wider area with dilation $r = 2$. (b) DeepMedic aligns and concatenates features from normal- and low-resolution inputs.

Attention U-Net. Shallow skip features contain background clutter as well as useful edges. **Attention U-Net** filters them with a **soft spatial attention gate** before concatenation: a semantically rich decoder signal g_i controls how much of the skip vector x_i^ℓ at voxel i is retained. After spatial alignment,

$$q_i^\ell = \psi^\top \mathrm{ReLU}(W_x^\top x_i^\ell + W_g^\top g_i + b_g) + b_\psi,$$
$$\alpha_i^\ell = \mathrm{sigmoid}(q_i^\ell), \qquad \widehat{x}_i^\ell = \alpha_i^\ell x_i^\ell.$$

The projections are $1 \times 1 \times 1$ convolutions; $\alpha_i^\ell \in (0, 1)$ is a scalar spatial weight shared by the channels of that voxel (fig. 35). Small weights suppress background features and attenuate the corresponding gradient flow. This learns where to focus without an external region-of-interest detector. Unlike Transformer self-attention, it gates features locally rather than modelling all pairwise spatial interactions.

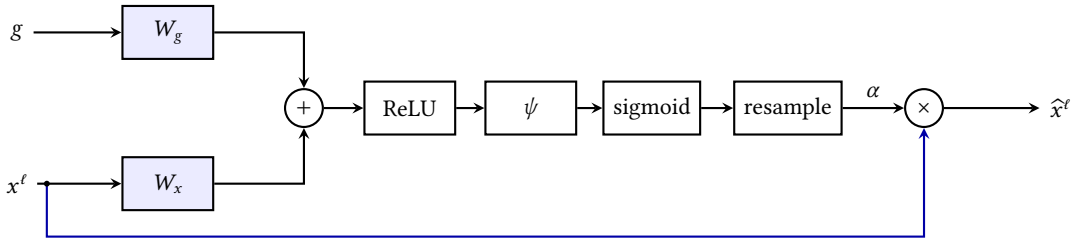


Figure 35: Attention gate: the decoder signal g and skip features x^ℓ produce spatial weights α; the weighted skip features then enter the decoder. Resampling aligns spatial grids.

5.3 Loss functions for medical image segmentation

Losses compare predicted class probabilities with reference labels at different levels: **distribution-based** losses act voxel by voxel; **region-based** losses measure overlap (Dice, IoU, Tversky); **boundary-based** losses account for distance to contours; and **topology-aware** losses target connectivity or holes, with

extra computational cost. Write $g_{i,c} \in \{0,1\}$ for the one-hot label and $p_{i,c}$ for the predicted probability of class c at voxel i, with N voxels and C classes.

Cross-entropy and weighting. The voxel-wise **cross-entropy** and **weighted cross-entropy** are

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} g_{i,c}\log p_{i,c},$$
$$\mathcal{L}_{\mathrm{WCE}} = -\frac{1}{N}\sum_{i=1}^{N} w(x_i)\sum_{c=1}^{C} w_c g_{i,c}\log p_{i,c}.$$

CE minimises the KL divergence from the reference labels to the predictions and provides smooth voxel-wise gradients, but a large background can dominate the loss. Class weights w_c (e.g. inverse class frequencies) rebalance classes; spatial weights $w(x_i)$ can emphasise borders. CE supports probability estimation, whereas Dice alone often gives poorly calibrated, overconfident outputs.

Focal loss. Static class weights do not distinguish easy background from ambiguous boundaries. If t_i is the true class at voxel i, **focal loss** uses the current prediction to downweight easy examples:

$$\mathcal{L}_{\mathrm{Focal}} = -\frac{1}{N}\sum_{i=1}^{N} \alpha_{t_i}(1-p_{i,t_i})^{\gamma}\log p_{i,t_i}.$$

Here α_{t_i} is a class weight and $\gamma \geq 0$ controls the focus, commonly $\gamma = 2$. As $p_{i,t_i} \to 1$, the modulating factor tends to zero for $\gamma > 0$, concentrating training on hard examples. A large γ can overemphasise noisy annotations.

Definition 5.1 (Soft Dice loss)**.** For one foreground class, with probabilities p_i and binary targets g_i, the **soft Dice loss** is

$$\mathcal{L}_{\mathrm{Dice}} = 1 - \frac{2\sum_{i=1}^{N} p_i g_i + \epsilon}{\sum_{i=1}^{N} p_i + \sum_{i=1}^{N} g_i + \epsilon}, \qquad \epsilon > 0.$$

It replaces the hard masks in the Dice coefficient (definition 4.1) with probabilities; ϵ stabilises the denominator.

True negatives do not contribute directly, making Dice useful for foreground/background imbalance. It nevertheless has noisy gradients for tiny or absent targets and is insensitive to the spatial arrangement of errors.

Definition 5.2 (Tversky loss)**.** With soft counts $\mathrm{TP} = \sum_i p_i g_i$, $\mathrm{FP} = \sum_i p_i(1-g_i)$ and $\mathrm{FN} = \sum_i (1-p_i)g_i$, the **Tversky loss** is

$$\mathcal{L}_{\mathrm{Tversky}} = 1 - \frac{\mathrm{TP}+\epsilon}{\mathrm{TP} + \alpha\,\mathrm{FP} + \beta\,\mathrm{FN} + \epsilon}, \qquad \alpha,\beta \geq 0, \quad \alpha+\beta = 1.$$

Equal weights $\alpha = \beta = 1/2$ give Dice overlap in the absence of smoothing; choosing $\beta > \alpha$ penalises false negatives more strongly, favouring sensitivity when missed lesions are costly. For multi-class imbalance, **generalised Dice** weights each non-empty reference class inversely by its squared volume, $w_c = 1/(\sum_i g_{i,c})^2$, so large organs do not dominate small ones.

Compound loss. Dice handles overlap and class imbalance, while CE supplies stable voxel-wise feedback. Their combination, used by nnU-Net, is

$$\mathcal{L}_{\mathrm{total}} = \lambda_{\mathrm{Dice}}\mathcal{L}_{\mathrm{Dice}} + \lambda_{\mathrm{CE}}\mathcal{L}_{\mathrm{CE}}, \qquad \text{typically } \lambda_{\mathrm{Dice}} = \lambda_{\mathrm{CE}} = 1.$$

Evaluation should pair DSC with surface distances such as HD and ASD (definition 4.2), since overlap alone misses boundary errors.

Boundary loss. Overlap does not distinguish equally sized errors near and far from the reference boundary. Let G be the reference foreground and $\phi_G(x)$ its **signed distance map**, negative inside G, positive outside and zero on ∂G. The **boundary loss** is

$$\mathcal{L}_{\text{Boundary}} = \sum_{i=1}^{N} \phi_G(x_i) p_i.$$

Minimisation favours foreground probabilities inside the reference and penalises false positives more strongly the farther they lie outside. This encourages accurate surfaces but is not itself HD or ASD. It is trained together with a region loss, gradually increasing its contribution, e.g. $\mathcal{L} = (1 - \alpha_t)\mathcal{L}_{\text{Dice}} + \alpha_t \mathcal{L}_{\text{Boundary}}$ as training progresses.

Topology and centreline Dice. A tiny break in a vessel or airway may barely change volumetric overlap while destroying connectivity. Let P, G be the predicted and reference foregrounds and S_P, S_G their skeletons. **Centreline Dice** (clDice) combines the fraction of the predicted skeleton contained in the reference with the fraction of the reference skeleton recovered by the prediction:

$$T_{\text{prec}} = \frac{|S_P \cap G|}{|S_P|}, \qquad T_{\text{sens}} = \frac{|S_G \cap P|}{|S_G|}, \qquad \mathcal{L}_{\text{clDice}} = 1 - \frac{2 T_{\text{prec}} T_{\text{sens}}}{T_{\text{prec}} + T_{\text{sens}}}.$$

For training, differentiable soft skeletonisation uses iterative morphological min/max pooling. Combining $\mathcal{L}_{\text{Dice}}$ with $\mathcal{L}_{\text{clDice}}$ encourages both correct vessel thickness and preserved connectivity.

Choosing a loss. Dice plus CE is a general baseline; focal plus Dice targets sparse lesions, asymmetric Tversky favours high sensitivity, a boundary term emphasises contour accuracy, and clDice is useful for tubular networks. These choices target different errors, so the evaluation must also reflect the intended task.

5.4 Transformer-based and general-purpose segmentation

From classification tokens to dense predictions. A segmentation Transformer keeps the spatial patch representations instead of reading only a classification token. In **SETR** (Segmentation Transformer), an image is divided into patches, linearly embedded with position information and processed by a pure Transformer encoder. A decoder reshapes the sequence to a spatial grid and upsamples to pixel labels, either progressively or by aggregating features from several encoder depths. Self-attention gives global context, while decoding recovers spatial resolution.

Hybrid CNN–Transformer designs. **TransUNet** uses a CNN to extract local, high-resolution features and a Transformer to encode global relations in the deepest features. A U-shaped decoder upsamples these representations and concatenates CNN skip features for precise localisation. Global attention over many 3D tokens is expensive, so **CoTr** sends multi-scale features from deeper CNN stages to a **deformable Transformer**: each query attends to a small set of sampled key positions instead of every spatial position. The reduced computation and memory permit attention over multi-scale features, including higher-resolution levels. The Transformer outputs are reshaped and supplied to corresponding decoder stages (fig. 36).

UTNet. **UTNet** inserts attention into both encoder and decoder stages of a CNN, modelling long-range relations at several scales. It reduces the keys and values spatially and uses relative position encoding, making attention approximately linear in the number of query positions when the sampled key grid is fixed. The attention decoder recovers detail from encoder skip features; the hybrid design can be trained without large-scale pre-training. In cardiac MRI, cross-vendor testing probes robustness to acquisition differences.

UNETR. **UNETR** applies a Transformer directly to linearly embedded 3D patches. Intermediate encoder outputs are reshaped and projected to different spatial scales, then passed through skips to a convolutional decoder. Unlike TransUNet, it does not rely on a CNN to produce the Transformer input features.

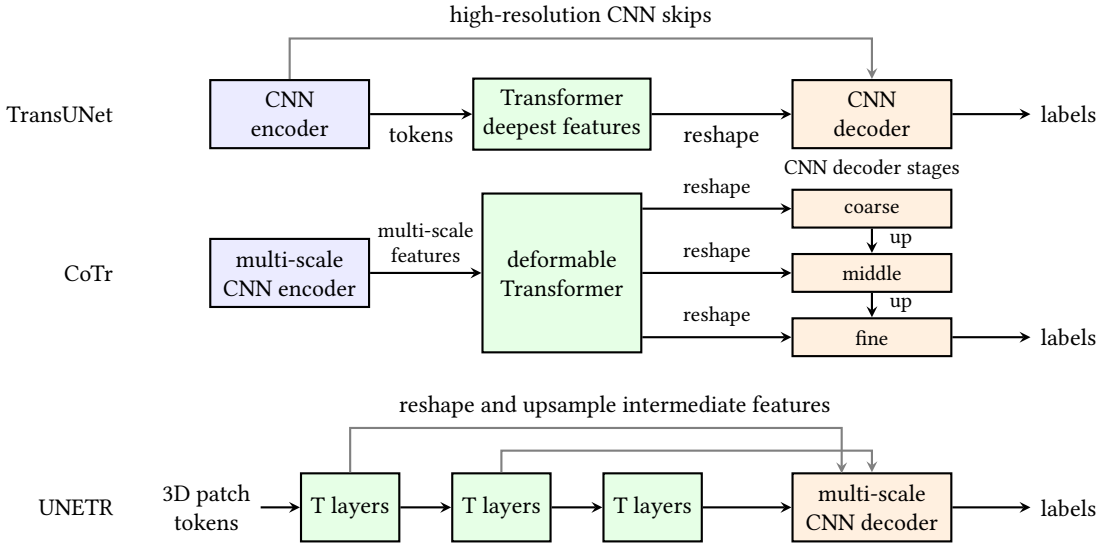


Figure 36: Decoder inputs: CNN skips in TransUNet, reshaped multi-scale Transformer outputs at several decoder stages in CoTr, and projected intermediate Transformer features in UNETR.

Specialist and general-purpose models. A model trained for a specific anatomy and modality does not automatically transfer to another task. Broad pre-training supports more reusable models, with two different interfaces: **promptable segmentation**, where a user indicates a target interactively, and **automatic multi-organ segmentation**, where the model returns a predefined set of anatomical labels without prompts.

Segment Anything Model (SAM). **SAM** is a promptable model pre-trained on natural images, with three components (fig. 37). A heavy ViT image encoder, pre-trained with a **masked autoencoder** (MAE), computes spatial image embeddings once; a prompt encoder represents points and boxes by positional embeddings and rough masks by convolutions; a lightweight mask decoder uses two-way attention between image and prompt representations to predict masks and estimated IoU scores. Reusing the image embeddings makes prompt updates cheap. An ambiguous click can return three candidate masks, corresponding for example to a whole object, a part or a subpart.

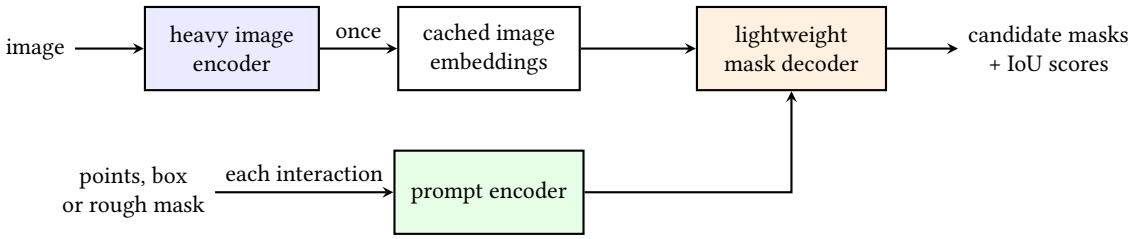


Figure 37: SAM separates image encoding from interactive prompting: cached image embeddings and encoded prompts feed a lightweight decoder that predicts candidate masks and quality scores.

Medical transfer and limitations. Zero-shot SAM works best when the target is well circumscribed and has clear contrast. Weak boundaries, infiltrative lesions and thin branching structures are difficult: prompts may cover mostly background or leak into neighbouring tissue. Independent 2D processing

also gives inconsistent masks between slices and can require repeated prompts. SAM identifies the indicated region, without automatically assigning a semantic anatomical label; semantic or text-driven selection needs an additional detector or text encoder.

MedSAM adapts SAM to heterogeneous medical images using bounding-box prompts, which are less ambiguous than a single point in low-contrast scans. **SAM 2** adds streaming memory attention for video: medical adaptations such as **MedSAM-2** treat a volume's ordered slices as frames, propagating segmentation from a few prompted slices through a memory bank.

Efficient adaptation. Full fine-tuning of a large image encoder is expensive and can cause catastrophic forgetting on a small local cohort. **Parameter-efficient fine-tuning** (PEFT) can freeze the encoder and train lightweight decoder components; **low-rank adaptation** (LoRA) is another strategy, using low-rank updates to the weights.

Automatic multi-organ segmentation. **TotalSegmentator** was developed for CT and later extended to MRI. It uses nnU-Net with broadly curated anatomical annotations to return multiple organ, bone, muscle and vessel labels automatically. It illustrates how a conventional 3D CNN and extensive, iteratively refined data curation can support many anatomical targets. Prompt-free inference is useful for large cohorts and automatic extraction of measurements such as body composition, organ volumes and vascular calcification.

Ambiguity and probabilistic segmentation. Expert contours can disagree because of partial volume effects or uncertain lesion margins (section 4). A single deterministic mask hides the range of plausible annotations. **Probabilistic U-Net** combines U-Net with a **conditional variational autoencoder** (cVAE): a Gaussian latent variable $z \sim \mathcal{N}(\mu, \Sigma)$ represents alternative plausible segmentations. At inference, an image-conditioned prior supplies samples of z, which are combined with the U-Net features to generate different masks (fig. 38). During training, a posterior also sees the reference annotation; the objective combines segmentation cross-entropy with a KL term aligning posterior and prior.

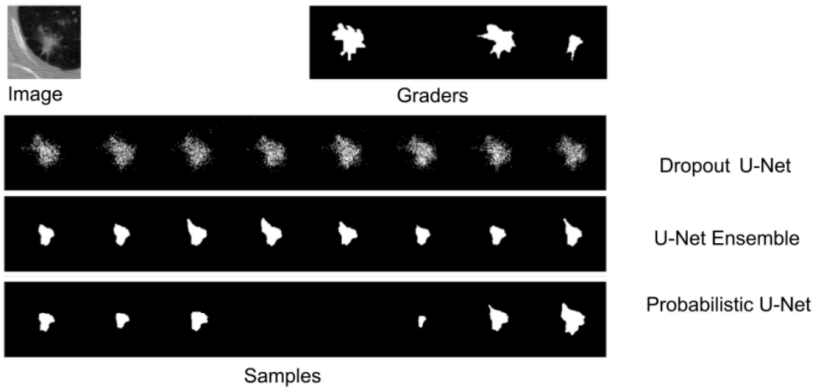


Figure 38: Expert masks of an ambiguous lesion (top) and model samples (bottom rows); probabilistic U-Net captures distinct shapes, including an empty mask. Source: Kohl et al., NeurIPS 2018.

6 Image Registration I

6.1 Registration and its applications

Definition. **Image registration** transforms two or more images into one common coordinate system. Its aim is to establish **spatial correspondences**: a transformation ϕ maps each point of one image to the anatomically corresponding point of the other, which amounts to deforming the underlying spatial

domain. The alignment sought can be geometric, anatomical or functional, e.g. to fuse CT, MRI and MR angiography of the same head into one view. Registration is **mono-modal** when the images come from the same modality (e.g. two T1-weighted MR images) and **multi-modal** otherwise (e.g. CT and MR).

Applications. Registration is used for intervention and treatment planning before an operation, e.g. by aligning MRI and CT of the same head; for computer-assisted surgery, by relating the patient during the operation to the pre-operative images; for computer-aided diagnosis, by overlaying functional PET on anatomical MRI; for disease monitoring at follow-up, e.g. the growth of a lung tumour in successive CT scans; for longitudinal growth analysis, e.g. of brain development; for motion analysis in dynamic images, e.g. cardiac MRI or 3D fetal ultrasound; and for population analysis across subjects.

Intra-patient registration aligns images of the same patient. Its applications are (multi-modal) image fusion, such as pre-interventional MRI with ultrasound acquired during brain-tumour surgery, or CT with PET for diagnosis; the combination of several images to enhance image quality, e.g. compounding of 4D ultrasound; the quantification of change; and motion analysis, e.g. of respiratory motion between inspiration and expiration or of cardiac motion. **Inter-patient registration** aligns images of different patients, pairwise or groupwise. Its applications are population analysis, the construction of **templates** (atlases) of a population or subpopulation, the modelling of the main variations within a population (e.g. shape modelling), and segmentation by registration.

Multi-atlas segmentation. An **atlas** is an image together with an expert segmentation of it. Registering the atlas image to a new image and applying the same transformation to the atlas labels segments the new image; this is **label propagation**. In **multi-atlas segmentation** a database of atlases is registered to the new image, giving one propagated segmentation per atlas, and **label fusion** combines them into the final segmentation. The simplest fusion rule is **majority voting**: each voxel receives the label chosen by most propagated segmentations. For example, if at a voxel two of five segmentations say hippocampus and three say background, the voxel is labelled background. The accuracy (Dice coefficient with respect to manual segmentations, definition 4.1) increases with the number of fused atlases, steeply at first and then levelling off.

Like ensemble learning, multi-atlas segmentation is robust and accurate; it yields plausible segmentations, implicitly takes several expert opinions into account and is fully automatic. On the other hand it is computationally expensive, cannot deal well with abnormalities such as tumours, and requires many labelled atlases.

6.2 Transformations and resampling

Components. Registration aligns a **moving image** M, defined on a domain Ω_M, to a **fixed image** F, defined on Ω_F. A **spatial transformation** $\phi : \Omega_F \to \Omega_M$ maps each point of the fixed domain to the corresponding point of the moving image, so the transformed moving image $M \circ \phi$, with $(M \circ \phi)(\mathbf{x}) = M(\phi(\mathbf{x}))$, lives on the fixed domain and can be compared with F point by point. The **transformation model** Φ is the space of admissible transformations, and an **interpolation** method evaluates M between its pixels. An **objective** (loss) function $\mathcal{J}$ scores a transformation by combining an image **similarity** (distance) measure $\mathcal{D}$ with a **regularisation term** $\mathcal{R}$, and an **optimisation** method searches for its minimum.

Definition 6.1 (Pairwise image registration)**.** Given F, M and Φ, find

$$\phi^* = \arg\min_{\phi \in \Phi} \mathcal{J}(\phi, F, M), \qquad \mathcal{J}(\phi, F, M) = \mathcal{D}(F, M \circ \phi) + \alpha\, \mathcal{R}(\phi),$$

where $\alpha \geq 0$ weights the regularisation.

The optimisation is iterative (algorithm 4) and usually multi-resolution (section 6.3).

Algorithm 4 Generic pairwise image registration

Require: fixed image F, moving image M, transformation model Φ, initial transformation $\phi^{(0)} \in \Phi$
Ensure: final transformation $\phi^* \in \Phi$

$i \leftarrow 0$
loop
  transform the moving image: $M \circ \phi^{(i)}$
  evaluate the objective $\mathcal{J}(\phi^{(i)}, F, M)$
  if good enough (stopping criterion) **then**
    return $\phi^* = \phi^{(i)}$
  end if
  update the transformation to $\phi^{(i+1)} \in \Phi$; $i \leftarrow i + 1$
end loop

Image geometry. An image is either a continuous function $F : \mathbb{R}^d \to \mathbb{R}$, with intensity $F(\mathbf{x})$ at $\mathbf{x} \in \Omega_F \subset \mathbb{R}^d$, or a discrete d-dimensional array $F \in \mathbb{R}^{m_1 \times \cdots \times m_d}$ on $\Omega_F = [0, m_1] \times \cdots \times [0, m_d]$. Its position in physical space is given by meta information: the **pixel** (voxel) **spacing**, the **origin** and the **direction** (orientation) of the image axes, which map index coordinates (x, y) to the **world coordinate system** (X, Y) (fig. 39). In 2D homogeneous coordinates

$$\begin{pmatrix} X \\ Y \\ 1 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & o_x \\ 0 & 1 & o_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} d_{xx} & d_{yx} & 0 \\ d_{xy} & d_{yy} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{T_{\mathrm{ItW}}} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} :$$

the indices are scaled by the spacing, rotated by the direction matrix, whose columns (d_{xx}, d_{xy}) and (d_{yx}, d_{yy}) are the world directions of the two image axes, and translated by the origin. The inverse $T_{\mathrm{WtI}} = T_{\mathrm{ItW}}^{-1}$ maps world to image coordinates.

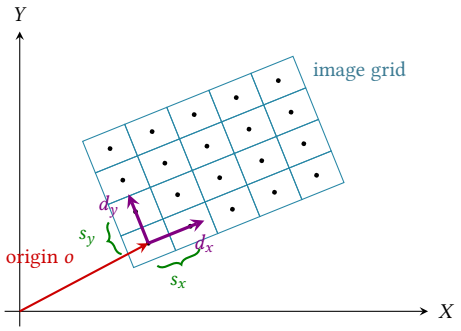


Figure 39: Image grid in world coordinates: origin o (red), directions d_x, d_y of the image axes (violet) and pixel spacing s_x, s_y (green); dots are pixel centres.

Registration is carried out in world coordinates, but there is more than one world: each image is placed in the world coordinates of its own acquisition, and two images brought into world coordinates are in general still misaligned. Registration estimates the transformation Φ_{BtA} from the world coordinates of image B to those of image A, so that index coordinates of B map to index coordinates of A by

$$\begin{pmatrix} x_A \\ y_A \\ 1 \end{pmatrix} = T_{\mathrm{WtI}}^{A}\, \Phi_{\mathrm{BtA}}\, T_{\mathrm{ItW}}^{B} \begin{pmatrix} x_B \\ y_B \\ 1 \end{pmatrix}.$$

Transformation models. The models are ordered by their number of **degrees of freedom** (DOF), i.e. of parameters (fig. 40). The **linear** models, linear in homogeneous coordinates, are the identity; **rigid**

transformations (rotation and translation; 3 DOF in 2D, 6 in 3D); **similarity** transformations (rigid plus uniform scaling; 4 and 7 DOF); and **affine** transformations (rigid plus non-uniform scaling and shear; 6 and 12 DOF). **Nonlinear** or **deformable** transformations have thousands to millions of parameters; poly-affine models and parameterised deformations lie in between.

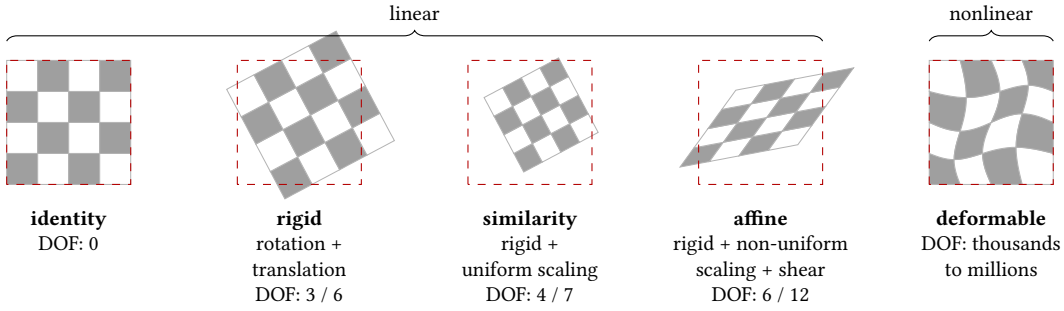


Figure 40: A checkerboard under the transformation models (dashed: original outline), with the degrees of freedom in 2D / 3D.

In 2D homogeneous coordinates an affine transformation is a product of a translation T, a rotation R by the angle θ, a scaling S and a matrix H with the shear angle ω:

$$T = \begin{pmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{pmatrix}, \quad R = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad S = \begin{pmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad H = \begin{pmatrix} \cos\omega & \sin\omega & 0 \\ -\sin\omega & \cos\omega & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

H has the form of a rotation: the shear is HSH^{-1}, which rotates the axes by ω, scales along them and rotates back, turning a square into a parallelogram (a rhombus for $\omega = 45^\circ$, fig. 41). Altogether

$$\Phi = HSH^{-1}RT = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{pmatrix},$$

with 6 DOF: translation (2), rotation (1), scaling (2) and shear (1).

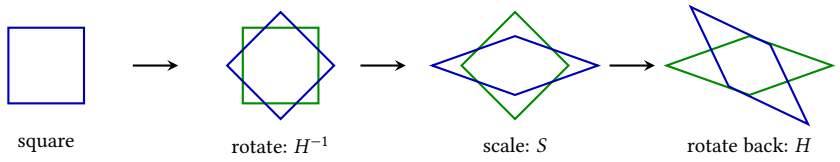


Figure 41: Shear as HSH^{-1} with $\omega = 45^\circ$, $s_x > 1 > s_y$: in each step the green shape is mapped to the blue one.

Deformable transformations. A deformation is described by its **displacement field** $\mathbf{u} = (u_1, \dots, u_d)$, $\phi(\mathbf{x}) = \mathbf{x} + \mathbf{u}(\mathbf{x})$, and visualised by the displacement in each direction or by a deformed grid. **Parametric** models describe $\mathbf{u}$ with few parameters: in spline or control-point models the displacements of a coarse grid of control points are interpolated by basis functions centred at the control points, e.g. B-splines, as in **free-form deformations** (FFD), which gives a low-dimensional deformation model. **Non-parametric** models estimate a **dense displacement field**, one displacement vector per pixel or voxel.

Forward and backward mapping. Transforming an image requires an intensity at every pixel of the output grid. **Forward mapping** sends each input pixel to the output: the mapped pixels fall between the output grid points, so some output pixels receive no value (gaps) and others several (fig. 42a). **Backward mapping** visits each output pixel instead and maps it back with the inverse transformation into the

input image, where it falls between pixels and its intensity is interpolated (fig. 42b). Every output pixel then gets exactly one value, so backward mapping prevents voxel gaps and is the rule for resampling. In registration it is built into the convention $\phi : \Omega_F \to \Omega_M$: for each grid point $\mathbf{x}$ of F, $(M \circ \phi)(\mathbf{x})$ is obtained by interpolating M at $\phi(\mathbf{x})$, which in general lies between the pixels of M.

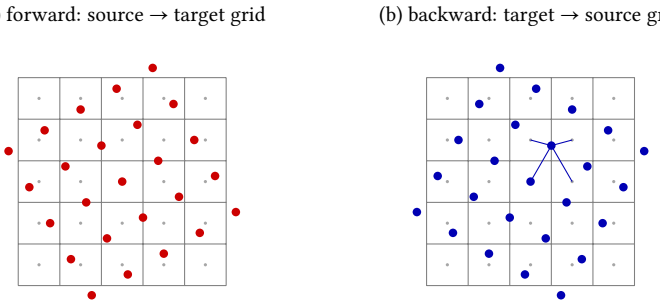


Figure 42: Rotation of a 5×5 image by 30° (grey dots: pixel centres of the grid drawn). (a) Forward mapping: source pixel centres (red) land between target pixels, leaving some target pixels empty and others with two values. (b) Backward mapping: each target pixel centre (blue) is mapped into the source grid and interpolated from the four surrounding source pixels (lines, shown for one).

Interpolation. In 1D, **nearest-neighbour** interpolation takes the value of the closest sample, **linear** interpolation joins neighbouring samples by straight lines, and **spline** interpolation fits a smooth piecewise polynomial (fig. 43a–c). In 2D, **bilinear** interpolation at a point p of the cell with corners p_0 (bottom left), p_1, p_2, p_3 and fractional offsets $r, s \in [0, 1]$ from p_0 (fig. 43d) gives

$$I(p) = (1 - r)(1 - s)\, I(p_0) + r(1 - s)\, I(p_1) + (1 - r)s\, I(p_2) + rs\, I(p_3),$$

and **trilinear** interpolation does the same in 3D with the eight corners of a voxel cell. Every resampling introduces errors (nearest neighbour produces blocky edges, linear interpolation blurs), so an image should not be resampled many times.

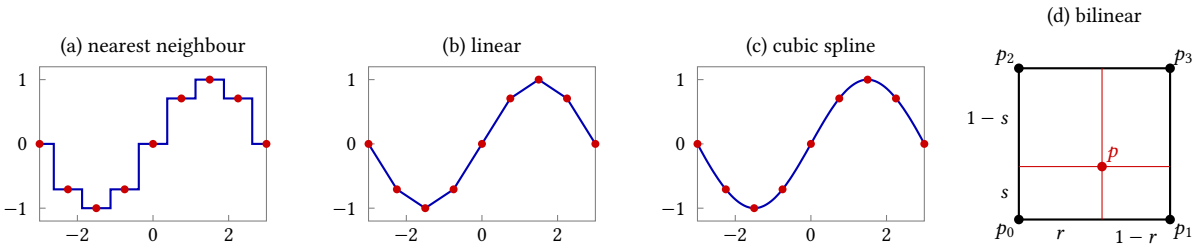


Figure 43: Interpolation of the samples of $\sin(\pi x/3)$ (red) by (a) nearest neighbour, (b) linear and (c) cubic-spline interpolation; (d) bilinear interpolation of the point p from the corners of its cell.

6.3 Similarity measures, regularisation and optimisation

Voxel-similarity measures are evaluated on the **overlap** Ω of the fixed image and the transformed moving image; below, $\mathbf{x}_1, \ldots, \mathbf{x}_N$ are the N grid points of F in Ω.

Mono-modal measures. The simplest assumption is an identity relationship between the intensities of the two images, which leads to the sum of squared differences; normalised cross correlation assumes a linear relationship.

Definition 6.2 (SSD and NCC)**.** The **sum of squared differences** and the **normalised cross correlation** are

$$\mathcal{D}_{\mathrm{SSD}}(F, M\circ\phi) = \frac{1}{N}\sum_{i=1}^{N}\big(F(\mathbf{x}_i) - (M\circ\phi)(\mathbf{x}_i)\big)^2,$$
$$\mathcal{D}_{\mathrm{NCC}}(F, M\circ\phi) = \frac{\sum_i \big(F(\mathbf{x}_i)-\mu_F\big)\big((M\circ\phi)(\mathbf{x}_i)-\mu_M\big)}{\sqrt{\sum_i\big(F(\mathbf{x}_i)-\mu_F\big)^2\,\sum_i\big((M\circ\phi)(\mathbf{x}_i)-\mu_M\big)^2}},$$

where μ_F and μ_M are the average intensities of F and $M\circ\phi$.

SSD (with the factor $1/N$ also called mean squared error, MSE) is the optimal measure if the images differ only by Gaussian noise, but it is sensitive to outliers and domain shifts; it is minimised. NCC is useful when the images were acquired with different intensity windowing; it is maximised. As $\mathcal{J}$ is minimised, a measure to be maximised enters it with a minus sign, e.g. $\mathcal{D} = -\mathcal{D}_{\mathrm{NCC}}$.

Multi-modal registration. For a T1-weighted brain slice and a rotated copy of it, SSD has a sharp minimum at the correct alignment (fig. 45a). Between modalities the identity assumption fails: the same tissue has different, non-linearly related intensities, e.g. cerebrospinal fluid is dark in T1- and bright in T2-weighted MRI, and such intensity inversions produce false minima of SSD (fig. 45b). SSD and MSE must therefore never be used across modalities.

Joint histograms. Multi-modal measures treat image histograms as probability distributions.

Definition 6.3 (Joint histogram)**.** The **joint histogram** $h(a, b)$ of two images counts the voxels of the overlap that have intensity a in the first image and intensity b in the second. The joint and marginal probabilities of the intensities are

$$p(a,b) = \frac{h(a,b)}{N}, \qquad p(a) = \sum_b p(a,b), \qquad p(b) = \sum_a p(a,b).$$

For identical images the joint histogram is the diagonal $a = b$, and the marginals are the histograms of the single images. For registered images of different modalities it shows compact clusters, one per tissue (e.g. air, skull, grey and white matter for MR and PET). Misregistration places different tissues at the same position, so the joint histogram disperses, and the more so the larger the misalignment (fig. 44).

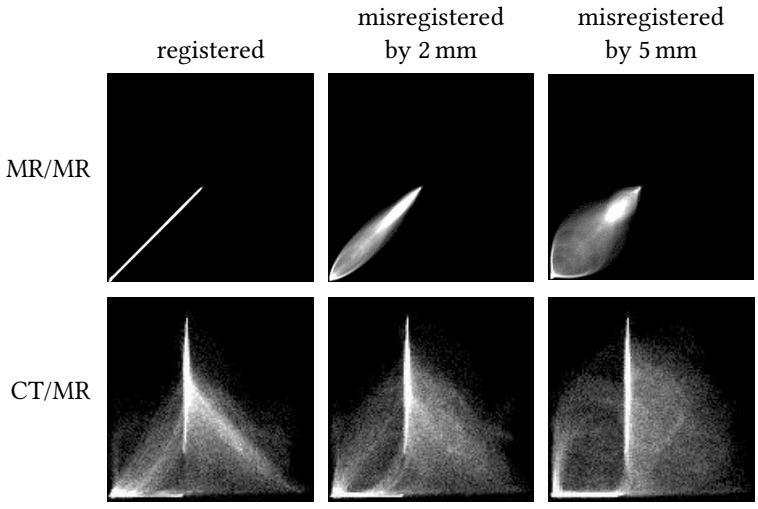


Figure 44: Joint histograms of two head images, MR/MR (top) and CT/MR (bottom), registered and misregistered (brightness: number of voxels).

Entropy and mutual information.

Definition 6.4 (Entropy and mutual information)**.** The **entropy** of an image A and the **joint entropy** of two images A, B are

$$H(A) = -\sum_a p(a) \log p(a), \qquad H(A,B) = -\sum_a \sum_b p(a,b) \log p(a,b).$$

The **mutual information** (MI) and the **normalised mutual information** (NMI) of F and $M \circ \phi$ are

$$\mathcal{D}_{\text{MI}}(F, M \circ \phi) = H(F) + H(M \circ \phi) - H(F, M \circ \phi), \qquad \mathcal{D}_{\text{NMI}}(F, M \circ \phi) = \frac{H(F) + H(M \circ \phi)}{H(F, M \circ \phi)}.$$

Entropy measures the amount of information, or disorder, in an image: it is maximal if all intensities are equally probable and minimal (zero) if one intensity has probability $p(a) = 1$. The joint entropy measures the information in the two images combined, and it grows as the joint histogram disperses. MI describes how well one image is explained by the other. Registration maximises MI or NMI, which clusters the joint histogram, and this works across modalities (fig. 45c).

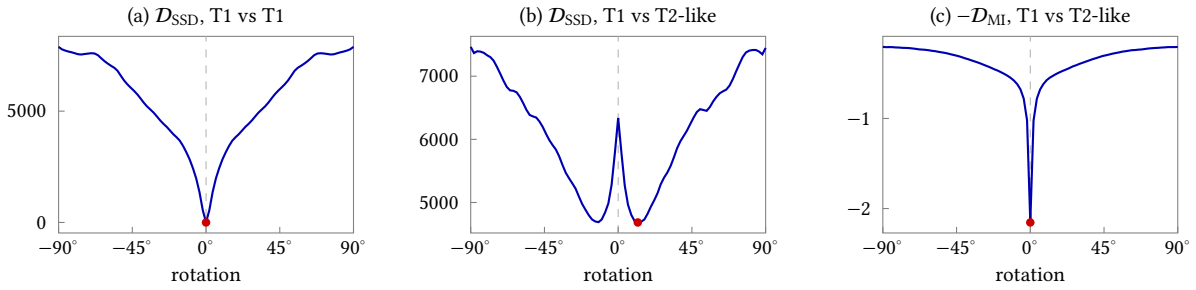


Figure 45: Distance between a fixed T1-weighted brain slice and a moving image rotated by the angle on the horizontal axis; red dots: minima. (a) SSD, moving image = the same slice. (b) SSD and (c) negative MI, moving image = a T2-like version of the slice (non-monotonic intensity map: cerebrospinal fluid bright, white matter dark).

MIND. Another way to handle multi-modal images is to convert the problem into a mono-modal one with the **modality independent neighbourhood descriptor** (MIND). For each pixel $\mathbf{x}_i$, a set of weights compares the image patch around $\mathbf{x}_i$ with the patches around the pixels $\mathbf{x}_j$ of a local search region R, using intensity differences within the same image:

$$w(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\sum_{\Delta\mathbf{x}} \| I(\mathbf{x}_i + \Delta\mathbf{x}) - I(\mathbf{x}_j + \Delta\mathbf{x}) \|^2}{\sqrt{2}\, \sigma^2} \right), \qquad \mathbf{x}_j \in R,$$

where $\Delta\mathbf{x}$ runs over the offsets within a patch and σ sets the scale. These weights describe the local structure around $\mathbf{x}_i$ rather than the intensities of the modality, so they can be compared directly between images of different modalities: the descriptors of corresponding points in MRI and CT are similar, and comparing them with the L^2 norm turns the problem into a mono-modal one.

Regularisation. Deformable registration is **ill-posed**: no closed-form solution exists, without further constraints there are infinitely many solutions, and they may be discontinuous or non-diffeomorphic (not one-to-one). The regularisation term $\alpha\mathcal{R}(\phi)$ of definition 6.1 makes the problem **well-posed**. In parametric registration the transformation model already restricts the search space; in non-parametric registration the regulariser constrains the problem itself. Its choice expresses which deformations are

plausible and realistic for the application, typically smooth ones. Two smoothness terms penalise the first or the second derivatives of the displacement field:

$$\mathcal{R}_{\text{diff}}(\phi) = \frac{1}{2}\int_{\Omega_F} \sum_{j,k=1}^{d} \left(\frac{\partial u_k}{\partial x_j}\right)^2 d\mathbf{x}, \qquad \mathcal{R}_{\text{bend}}(\phi) = \frac{1}{2}\int_{\Omega_F} \sum_{i,j,k=1}^{d} \left(\frac{\partial^2 u_k}{\partial x_j\, \partial x_i}\right)^2 d\mathbf{x},$$

the **diffusion** regularisation and the **bending energy**. Without regularisation ($\alpha = 0$) the transformed moving image matches the fixed one closely, but through a folded, implausible deformation; with $\alpha > 0$ the match is similar and the deformation smooth (fig. 46). Regularisation thus prevents folding and tearing, and α sets the trade-off between similarity and regularisation.

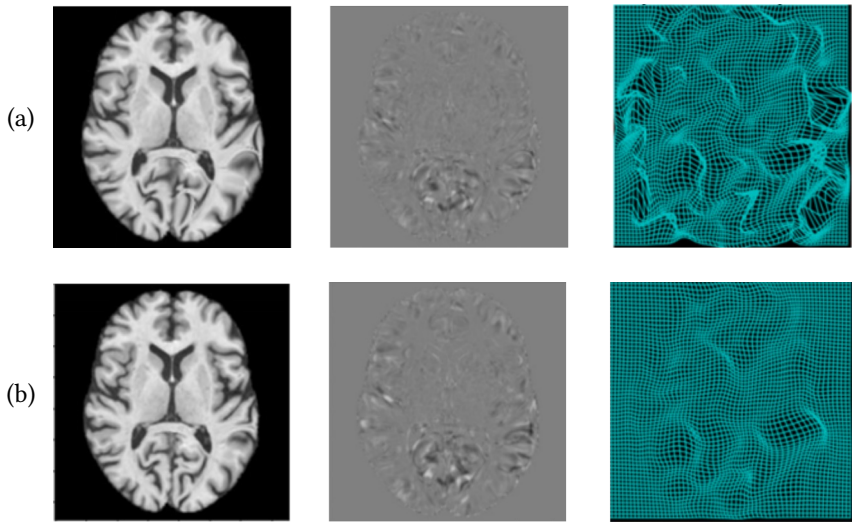


Figure 46: Deformable registration of two T1-weighted brain slices (a) without regularisation, $\alpha = 0$, and (b) with regularisation, $\alpha > 0$: transformed moving image, difference from the fixed image, deformed grid.

Optimisation. $\mathcal{J}$ can be minimised by gradient descent, stochastic, Bayesian, discrete or convex optimisation, the downhill simplex method and others.

Definition 6.5 (Global and local minimiser)**.** A point $\mathbf{x}^* \in \mathbb{R}^n$ is a **global minimiser** of f if $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all $\mathbf{x}$, and a **local minimiser** if there is a neighbourhood $\mathcal{N}$ of $\mathbf{x}^*$ such that $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \mathcal{N}$.

Registration wants the global minimiser; a local one may correspond to a partial alignment of the images. In one dimension a minimiser of a smooth f satisfies $f'(x^*) = 0$, as maxima also do, and in addition $f''(x^*) \geq 0$. The optimisation runs over the transformation space: n is the number of transformation parameters, $n = d \times$ (number of pixels) for a dense displacement field.

Multi-resolution optimisation. **Multi-scale** techniques register blurred versions of the images first, which increases the capture range for features that are far apart. **Multi-resolution** techniques also downsample the blurred images, which further increases robustness and speed. A **Gaussian pyramid** holds both images at several resolutions (e.g. 1/4, 1/2 and full resolution): registration starts at the coarsest level, each result initialises the next finer level, and the final registration is at full resolution; at every level the loop of algorithm 4 is run. This coarse-to-fine optimisation reduces the risk of getting trapped in local minima.

Choosing the components. Successful registration matches the transformation model and the similarity measure to the imaging physics and to the imaged anatomy and physiology, with an appropriate regularisation. Table 2 lists typical choices.

Scenario	Similarity measure	Transformation model	Key constraint or regulariser
multi-modal brain, CT to MRI (intra-subject)	MI or NMI	rigid (6 DOF)	the rigid skull preserves Euclidean distances
lung CT, inspiration to expiration (intra-subject)	SSD or NCC (mono-modal)	deformable (B-spline FFD or dense)	diffusion or bending energy, $\alpha > 0$
multi-modal abdomen, MRI to CT (intra-subject)	MIND with L^2 norm, or NMI	rigid, then deformable	diffusion or bending energy, $\alpha > 0$
MR atlas propagation (inter-subject)	NCC or MI	affine, then deformable (B-spline FFD or dense)	majority voting for label fusion

Table 2: Typical choices of registration components.